

Target-correlation maps are library-conditional but recoverable from a few hundred ligands in two large Vina panels

Adeliya Leleytner^{1,*}, Victor Safronov¹, Maxim Fedorov¹

¹Institute for Information Transmission Problems of the Russian Academy of Sciences

(Kharkevich Institute), Bolshoy Karetny per. 19, Moscow 127051, Russia

*Correspondence: a.leleytner@gmail.com

Abstract

Background. Dense docking matrices conflate spectral redundancy, target–target correlation maps and compound-level ranking.

Results. Two Vina matrices (12,651 ligands by 44 targets and 260,060 by 58) were dominated by a nearly uniform ligand-wide axis; within-ligand centring exposed residual maps with PR effective dimensions 9.303 and 18.206 for correlation spectra and 6.911 and 8.235 for covariance spectra; only the increase direction was robust across estimands. The maps were library conditional: in post-hoc within-library splits, low- versus high-molecular-weight maps agreed at only 0.205 and 0.351, versus 0.991 and 0.994 for molecular-weight-matched chemically disjoint controls. In support-specific sampling, 200 same-distribution ligands recovered source-library maps at 0.942 and 0.918. Physicochemical and count-Morgan bases predicted similar score structure out of chemical group, whereas a stable-hash component with mean held-out target R^2 near zero still resembled experimental correlation maps because column correlation removed its scale. Against a fixed centred endpoint from four published kinase panels, docking-side centring increments of 0.001–0.045 were unresolved; against uncentred endpoints they were 0.168–0.264. Subtracting a ligand’s row mean is exactly invariant to its target order, while unequal target scaling converts that term into a rank-one correction. On 6,480 ligands and 25,214 non-tied observed within-ligand target comparisons, absolute Vina scored 0.537 versus 0.529 for scaled residuals; an endpoint-restricted human binding K_i/K_d arm gave 0.559 versus 0.520, with chemical and target-composition intervals spanning zero. Its 0.013 advantage over a target-only prior fitted on

the external DOCKSTRING reference was likewise unresolved.

Conclusions. In the tested Vina systems, a target-correlation map is an observable of the scoring pipeline, receptor panel and observed source library: inexpensive to estimate on a support-specific representative sample, poorly portable across the tested molecular-size domains, and not a compound-level ranking rule. Residual dimension is not a stand-alone docking-quality criterion.

Scientific Contribution. We quantify molecular-size-domain non-transport and support-specific recovery of docking target- correlation maps, and identify a scale-free nuisance-projection failure in validating fitted component maps. We also make explicit that within-ligand centring cannot alter target order and identify unequal target scaling as the operation that breaks this invariance.

Keywords: molecular docking; AutoDock Vina; scoring-function bias; target-correlation structure; effective dimension; score normalization; kinase selectivity; reverse docking.

1 Introduction

A docking calculation produces one score for one ligand–target pose, but many cheminformatics applications use a matrix. Reverse docking compares targets for a compound, secondary-pharmacology panels search for off-target liabilities, and multiobjective design optimizes activity profiles rather than a single interaction. In these settings the object of interest is a matrix $X \in \mathbb{R}^{N \times P}$, with ligands in rows and targets in columns. Treating its P columns as P independent descriptors is unwarranted when a small number of shared score tendencies dominates the matrix.

Those three applications turn on a compound’s target *order*, and we show below that the most common correction cannot change it. The object that a docking panel does support is the target-pair map: which score columns co-vary across a stated ligand library. That map has real uses—diagnosing how redundant a screening panel is, choosing which targets a new panel should contain, and comparing two scoring set-ups on a shared ligand support—and this article is about how far it can be trusted, and about the sharp line between it and compound-level ranking.

Keeping those two apart requires being explicit about what is being estimated. We use *estimand* in the sense of ICH E9(R1) [43]: the precise specification of the quantity under estimation—here, jointly, which score transformation, which experimental endpoint, which ligand support and which target panel. Two analyses that differ in any one of these components

estimate different things even when they report the same summary statistic, and Table 1 fixes the terms this article uses. The practical consequence recurs throughout: an apparent benefit of normalizing docking scores can be produced entirely by transforming the experimental endpoint at the same time.

Two sources of shared variation have long been recognized. Empirical scoring functions can reward additional ligand atoms and contacts, so molecular size and flexibility affect scores across many targets [5, 7, 8]. Protein pockets also differ in size, contact area and composition, producing target-specific score offsets. Such effects motivated several ligand-wise, target-wise and bidirectional score normalizations for virtual screening and target fishing [6, 7, 15–18, 20]. Following the two-factor chemical–protein-interactome model of Yang et al., we write

$$X_{ij} = \mu + a_i + b_j + \gamma_{ij}, \quad (1)$$

where a_i is a ligand-wide favourable-score tendency, b_j is a target offset, and γ_{ij} is the remaining ligand–target term. The spectrum of the full matrix may be dominated by a_i even when γ_{ij} contains several correlated directions.

Matrix-wide correction and multivariate analysis of docking panels predate this work. MASC standardized each ligand’s scores across multiple active sites, while later studies showed that its enrichment effect depends on the reference target panel and scoring function. Fukunishi et al. derived receptor- and compound-similarity from a multi-receptor by multi-ligand docking-affinity matrix and subsequently applied PCA to a protein–compound interaction matrix. SePreSA and DRAR-CPI used sequential drug- and protein-direction Z transformations, while combined ligand- and target-score normalization was also used operationally [6, 7, 18–25].

These procedures are related to, but not algebraically identical with, the primary estimand used here. MASC additionally divides every ligand row by its own standard deviation, whereas 2DIZ sequentially standardizes in both matrix directions; our residual target-pair geometry subtracts ligand means and subsequently gives target columns unit variance. For any ligand with positive row standard deviation, MASC row standardization is a positive affine transformation across its targets and therefore preserves that ligand’s target order. Because the shift and scale are ligand-specific, it can nevertheless change the ordering of ligands within any one target. We therefore claim neither normalization nor docking-profile PCA as new. The contribution is to quantify how target-correlation maps change with chemical support, establish how many

representative ligands recover a source-library map, characterize a low-dimensional ligand-feature response within the residual, and distinguish map geometry from ligand-wise ranking.

Related work reinforces this distinction from two directions. Random-matrix methods have used spectral structure to regularize chemical-feature correlations for affinity prediction, whereas kinase-selectivity metrics can reorder compounds when the operational definition is changed [28–30]. Here PR is therefore used as a transparent dependence scale, not introduced as a new statistic; the methodological contribution is the complete audit that keeps matrix redundancy, target-pair co-response and within-ligand ranking separate.

The same distinction applies to experimental validation. Let $G(X)$ denote the target-by-target correlation matrix calculated across ligands. Comparing $G(X)$ with an experimental matrix $G(E)$ asks whether pairs of targets co-vary similarly across two libraries. Comparing a centred docking geometry $G_R(X)$ with a centred experimental geometry $G_R(E)$ changes both predictor and endpoint. A difference between those two matched comparisons cannot be assigned uniquely to transforming the docking scores. It is also not a test of whether the more active target is ranked correctly for an individual compound.

Two public, dense Vina-family matrices make it possible to quantify these distinctions at scale. DOCKSTRING contains scores for more than 260,000 molecules against 58 medically relevant targets [3]. A separately generated secondary-pharmacology matrix contains 12,651 analyzable compounds and 44 targets [4]. Four published kinase profiling resources—DAVIS, PKIS2, PKIS1 and KiRHub—provide dense experimental measurements on a common set of 20 kinases [31, 33–35].

The original DOCKSTRING report documented correlations among its target score columns [3]. Our analysis starts from that observation but asks different questions: whether a complete target-correlation map transports across ligand supports, how many same-support ligands recover it, whether fitted-component maps survive nuisance controls, and whether any matrix transformation improves ligand-wise target ordering.

We use these resources to separate four questions. First, what dependence remains after removal of ligand main effects? Second, does that target-correlation map transport between chemical domains, and how many representative ligands are needed to estimate it? Third, what low-dimensional ligand-feature response organizes the residual map, and what can agreement with fixed experimental co-response endpoints license? Fourth, does the same processing improve ligand-wise target order?

Table 1: Terms used in this article. Each row names one object, states it in symbols for a score matrix $X \in \mathbb{R}^{N \times P}$, and says what it is not.

Term	Definition	Distinct from
Estimand	The joint specification of score transformation, experimental endpoint, ligand support and target panel [43].	A statistic. Two analyses can share a statistic and estimate different quantities.
Within-ligand centring	$R_{ij} = X_{ij} - \bar{X}_i$: subtract each ligand’s mean over the target panel.	Target-wise centring or scaling, which act on columns.
Target-pair geometry	$G(X)$, the $P(P - 1)/2$ upper-triangle entries of the target correlation matrix across ligands.	A biological interaction network; it is conditional on the ligand library.
Participation ratio (PR)	$(\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ for correlation eigenvalues λ_k .	An algebraic rank, an independent statistic, or a docking-quality score.
Transformation-matched null	A null in which the same transformations are applied to permuted data.	A target-label null, which asks a different question.
Fitted-component map	The target-correlation map of an out-of-fold predicted surface.	Held-out predictive performance: column correlation can preserve this map even when held-out R^2 is zero because it removes the predicted columns’ scale.
Level and increment	A level is one geometry–endpoint concordance; an increment is a paired difference between two levels evaluated against the same fixed endpoint and support.	A raw/raw-to-centred/centred change, which transforms predictor and endpoint together.
Path dependence	Transforming predictor and endpoint in a different order gives a different attribution.	An interaction in the statistical-model sense; it is an ordering artefact.
Scaled two-way residual	$T_{ij} = (X_{ij} - \bar{X}_i - \mu_j + \bar{X}_{\text{ref}}) / s_j$ for reference target mean μ_j , reference grand mean $\bar{X}_{\text{ref}}$ and residual SD s_j .	Within-ligand centring alone, which cannot reorder targets.
Resolved (of a contrast)	Shorthand used throughout for: the stated interval excludes zero, or the stated permutation probability is below 0.05.	A claim of practical importance, and—for a negative—evidence of absence rather than absence of evidence.

The principal empirical finding answers the second question: in exploratory post hoc splits within the two observed source libraries, low- and high-molecular-weight maps disagree sharply despite near-ceiling agreement in molecular-weight-matched disjoint controls. Descriptive same-distribution experiments show high recovery of each source-library map from 200–500 ligands; they do not define a universal sample-size or accuracy threshold. The exact operational finding answers the fourth: subtracting a ligand’s row mean cannot change its target order, and only unequal target scaling turns that otherwise invariant term into a rank-one ranking correction. The spectral and feature decompositions explain these boundaries; they are not introduced as new PCA, centring or effective-rank estimators.

2 Methods

2.1 Study design and analysis status

The study separates three estimands: the target-correlation spectrum of a dense score matrix, the upper-triangular target-pair geometry across a stated ligand support, and within-ligand observed-pair target concordance. Analyses of the two large score matrices and the matched ChEMBL blocks were developed first. The fixed-20 experimental geometry, structural controls, centering-panel comparison and sampling-recovery analyses are exploratory post hoc analyses added during manuscript revision. The dense DAVIS and PKIS2 exact-match ranking checks, descriptor decomposition and molecular-size-domain splits were also added during revision and were not preregistered. Molecular weight was selected as the domain stratifier after preliminary inspection. KiRHub was inspected only after the 15-pair experimental endpoint had been fixed from DAVIS, PKIS2 and PKIS1, but this sequence was not preregistered. Statistical probabilities are therefore interpreted as calibrated descriptive evidence rather than confirmatory error control over the entire analysis history.

2.2 Large Vina score matrices

Docking-44 was derived from the secondary-pharmacology resource of Nikitin et al. [4]. It contains 12,651 analyzable compounds and 44 receptor columns scored with Vina-GPU 2.0 [2]. The upstream workflow replaced positive scores with zero before release. Accordingly, the frozen analyzable export contained no value above zero (111 cells were exactly zero), and reapplying the upper-zero rule made no further change. Its 8,159 missing cells (1.466%) were filled with the mean of their target column for the primary complete-matrix analysis. Missingness-aware and complete-case alternatives were analyzed separately.

DOCKSTRING v1 contains 260,155 molecule rows and 58 target-score columns [3]. The source table had 339 missing scores (0.00225%) across 95 rows. Because listwise deletion removed only 0.0365% of molecules, DOCKSTRING-58 used the 260,060 rows complete across all targets, without imputation; target-mean and target-median sensitivities retained every row. Scores were generated with the DOCKSTRING AutoDock Vina workflow [1]. Of 5,413 positive source scores, 5,393 occurred in retained rows and were set to zero in the primary analysis; retaining them was a sensitivity. The local data manifest records the score matrix and derived molecular identity contract; receptor files and target configurations remain available in the

checksum-anchored upstream DOCKSTRING release.

Cross-panel chemical overlap was calculated from recomputed Standard InChIKeys, 14-character connectivity blocks and canonical non-empty Bemis–Murcko scaffolds [46]. Target identities were curated by protein accession, and receptor files were compared by structure identifier and chain. These counts characterize overlap; they were not used to remove rows from the full spectral analyses.

2.3 Experimental kinase panels

All kinase-geometry analyses used the same ordered 20 targets: ABL1, AKT1, AKT2, CDK2, CSF1R, EGFR, FGFR1, IGF1R, JAK2, KDR, KIT, LCK, MAP2K1, MAPK1, MAPK14, MAPKAPK2, MET, PLK1, ROCK1 and SRC. DAVIS contributed a dense 72-by-20 pKd block from the CC0 DAVIS-Complete V3 export, which also supplies the compound structures and curated target sequences used here [32]. The affinities originate from Davis et al. [31]. The released 10,000-nM Kd upper bound becomes a pKd floor of 5: 965 cells (67.0%) lie at that limit, with 33.3–97.2% per target and 4 targets above 90%. PKIS2 contributed 645 compounds measured by single-dose KINOMEScan percentage displacement or inhibition at 1 μ M [34]. PKIS1 contributed 360 compounds measured by Nanosyn percentage inhibition at 1 μ M; duplicate compound identifiers were median-aggregated and the CDK2/cyclin A construct was primary [33]. KiRHub contributed a dense 92-by-20 residual-activity block measured at 1 μ M at the ATP K_m ; CDK2/cyclin A was primary [35]. Correlation geometry is invariant to reversing the orientation of all columns in a panel, so the pKd, inhibition and residual-activity directions do not create a sign ambiguity in the target-pair comparisons.

The DAVIS censoring sensitivity replaced each pKd value by whether it lay above the pKd-5 floor, and separately retained only targets with at least 5, 10 or 15 values above that floor. It also rebuilt the three-panel 15-pair endpoint after replacing continuous DAVIS correlation by binary above-floor-status correlation, then evaluated the resulting labels against centred KiRHub geometry. These analyses test robustness to the measurement limit; they do not recover quantitative affinity information below it.

To prevent direct ligand reuse in the docking reference, Standard-InChI connectivity blocks from DAVIS, PKIS2 and PKIS1 were unioned before extracting the 20-target DOCKSTRING matrix. The 1,050 excluded blocks removed 481 DOCKSTRING rows, leaving 259,579 reference ligands. An additional cyclic-Murcko exclusion and five fixed 15,000-ligand reference supports were

sensitivities. KiRHub source tables contain compound names but no structures, so KiRHub-specific molecular-identity exclusion was not possible.

Overlap among the three structure-annotated experimental panels was audited at three levels: full Standard InChIKey, its 14-character connectivity block and canonical non-empty Bemis-Murcko scaffold. Empty scaffolds were excluded from cross-panel scaffold intersections. Because KiRHub reports compound names without a frozen structure mapping, only exact normalized-name overlap with DAVIS was evaluated; PKIS project identifiers were not treated as comparable clinical names.

2.4 Centering, correlation spectra and effective dimension

For a complete matrix X , fitting the additive structure in Eq. 1 by least squares gives the two-way-centred residual

$$\Gamma_{ij} = X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{...} \quad (2)$$

The primary spectrum is the eigenspectrum of the target correlation matrix, which gives each target unit variance. Because target correlation is invariant to column offsets, correlating Γ is identical to subtracting only $\bar{X}_{i.}$ before calculating target correlations. We call this operation within-ligand centering and use full two-way centering when discussing the score-scale residual matrix. Correlation-spectrum PR was calculated by Eq. 6. Entropy effective rank, PC1 fraction and the number of components required for 90% variance were secondary summaries [41, 42].

Covariance-spectrum PR retained target-specific variances. Docking-44 missingness sensitivities used target-mean and target-median imputation, complete rows, and an additive least-squares fit using observed cells only; missing residuals were set to zero after fitting row and column effects in the latter analysis. Positive-score clipping was removed in a DOCKSTRING sensitivity.

The row-norm-preserving null sampled independent Gaussian directions in the $(P - 1)$ -dimensional target-zero-sum subspace and rescaled every simulated row to its observed residual Euclidean norm before column standardization. Five hundred draws were evaluated on fixed 12,000-ligand supports. Additive Gaussian, empirical target-column permutation and transformation-matched parallel-analysis nulls are reported in Supplementary Information. All apply the same centering and standardization operator to observed and null matrices.

2.5 Physicochemical slopes and residual-PC1 interpretation

RDKit molecular weight, heavy-atom count and rotatable-bond count were calculated from the analysis SMILES. Docking-44 used all 12,651 mean-imputed ligands. For DOCKSTRING, descriptor analysis used a score-blind simple-random sample of 15,000 complete ligands selected with NumPy seed 71; the full score matrix was not required for this interpretation. For each target we fitted

$$X_{ij} = \beta_{0j} + \beta_{Mj}(\text{MW}_i/100) + \beta_{Rj}\text{RotB}_i + \varepsilon_{ij}. \quad (3)$$

We report the distributions of β_{Mj} and β_{Rj} , the cosine between the target-wise molecular-weight residual-slope vector and residual-PC1 loadings, and the absolute correlation between molecular weight and residual-PC1 ligand scores. Component signs were oriented only for reporting. A Docking-44 sensitivity removed 4mqs, which contained 3,352 of the 8,159 imputed cells. RDKit rotatable bonds are a molecular proxy and were not treated as a decomposition of Vina’s internal torsional term.

The chemical-group-held-out predictive decomposition used seven descriptors in fixed order: heavy-atom count, molecular weight, Labute ASA, TPSA, cLogP, rotatable-bond count and ring count. Five-fold GroupKfold split the frozen Docking-44 Butina groups or DOCKSTRING Murcko groups; each acyclic DOCKSTRING molecule was a singleton group. Target offsets, residual target scales, descriptor means and scales, and the target-specific linear coefficients were re-estimated within every training fold. We pooled the out-of-fold residual score errors, recalculated their target correlations and PR, and compared mean squared target correlation before and after descriptor removal. Residual-PC1 R^2 used pooled descriptive loadings, but every ligand-level prediction remained group held out. This analysis measures predictable score structure and has no causal interpretation.

2.6 Cross-scorer sensitivity on fixed Vina-selected poses

To test whether the shared-axis pattern persists when the same poses are evaluated by other functions, we held the released DOCKSTRING Vina-selected top poses fixed and changed only the scoring function. A simple random subset of 2,000 ligands (seed 20260802) from the frozen 15,000-ligand support was taken across the nine targets whose pose archives are published, giving 18,000 cells. Poses were scored with smina using the Vina and Vinardo functions, and with ODDT implementations of RF-Score v1–v3, NNScore 2.0 and a pretrained linear PLECScore; RF-Score

and NNScore models were trained offline on the bundled PDBbind 2016 descriptor tables. Scores were not clipped, because clipping only the Vina-family columns would be asymmetric across scorers. For each function we computed the raw and within-ligand-centred correlation spectra, PR, PC1 fraction, mean squared off-diagonal correlation and the cosine of the PC1 loading vector with the uniform target direction, with 95% cluster-bootstrap intervals. The four criteria were stated before the non-Vina scores were inspected, but that ordering is not independently verifiable from the released package and the criteria should be read as a descriptive checklist rather than a preregistered test. This is a rescoring of retained Vina poses, not an independent re-docking: a function that would select different poses during its own search is not simulated, so the analysis estimates scorer sensitivity conditional on Vina pose selection. RF-Score v3 additionally consumes Vina terms as input features and is therefore not function-independent, and all three machine-learned scorers are trained on crystal complexes and applied off-distribution to docked poses. These criteria are weak: a broadly positive score matrix can have a near-uniform leading vector without establishing a chemically useful shared axis. The secondary ODDT stage ran under a separate Python 3.9/ODDT 0.7 environment; its released outputs are checksum-verified but are not recreated by the default Python 3.12 container or make all target.

2.7 Broad DOCKSTRING-ChEMBL ligand-wise benchmark

ChEMBL activities for the 31 human single-protein targets mapped to DOCKSTRING were refetched without a potency floor or assay-type restriction. The broad primary contract required non-null pChEMBL, exact standard relation and endpoint Ki, Kd, IC50 or EC50. The source pull contained 212,254 records; endpoint filtering retained 196,889, and removal of 146 records with unparseable structures left 196,743 for aggregation. Ligands were matched to DOCKSTRING by RDKit standard full InChIKey, repeated records were median-aggregated per ligand-target cell, and ligands observed on at least two targets were retained. A less assay-heterogeneous, endpoint-restricted sensitivity required a human binding assay and endpoint Ki or Kd. Its within-ligand comparisons may still pair a Ki-derived cell with a Kd-derived cell. Separate Ki, Kd, IC50 and EC50 arms exposed rather than averaged endpoint heterogeneity; no target pair in those arms mixes endpoint types.

Every docking transformation was fitted outside the complete evaluated ligand set. Target means, target standard deviations, residual target standard deviations and the grand mean came from the remaining complete DOCKSTRING reference; the evaluated ligand's row mean

used all 58 docking targets. Representations were absolute Vina, target-centred unscaled, two-way-centred unscaled, column-standardized, target-centred residual-scaled and scaled two-way residual scores. A docking target prior repeated the reference target means. An external ChEMBL prior used target means after excluding every evaluated ligand, and a benchmark-fitted cohort prior recomputed target means after excluding the held-out ligand’s entire Bemis–Murcko cluster, falling back to the external prior when necessary. The last is a diagnostic of assay-selection and target-prevalence structure, not a deployable ligand-specific predictor.

For every ligand, all experimentally observed unordered target pairs were compared; an experimental tie was excluded and a docking-score tie received half credit. Mean concordance gave each evaluated ligand equal weight. Paired differences were resampled 5,000 times by ligand, intact Murcko scaffold and intact Butina cluster. Butina clustering used radius-2, 2,048-bit Morgan fingerprints and Tanimoto 0.65; the conservative chemical interval is the union of the Murcko and Butina intervals. Delete-one-target jackknifing masked one experimental target at a time and reevaluated the already externally fitted score representations; it is reported separately as an evaluation-panel composition sensitivity. Chemical intervals remain conditional on the fixed target labels, while the jackknife is not a population confidence interval for all proteins.

2.8 Rank-matched ligand-feature and nuisance controls

We repeated the group-held-out decomposition on the DOCKSTRING reference filtered to remove structural overlap with the three structure-annotated panels and used for the fixed-panel comparison (259,579 ligands after excluding every connectivity block occurring in DAVIS, PKIS2 or PKIS1) and the same ordered 20 kinases. Every arm used the identical five scaffold folds, fold-local target offsets and scales, feature standardization and separate least-squares coefficient vector for each target. Predictive performance was summarized by mean out-of-fold target R^2 and by the operational reduction in mean squared target correlation after subtracting the fixed out-of-fold prediction.

All predictive bases had seven dimensions. The primary physicochemical basis contained the seven descriptors above. The capacity-matched chemical control contained one total Morgan-environment-count coordinate and six random projections of unnormalised radius-2, 2,048-bin Morgan count fingerprints. The first of 20 consecutively seeded projections was locked as the representative before their performance was inspected; the full seed range is an algorithmic sensitivity, not an uncertainty interval. Two unit-normalised bit-fingerprint projections tested

the effect of removing count magnitude. A stable nuisance basis mapped each exact input SMILES deterministically to seven centred BLAKE2b-derived coordinates. Twenty seeded row permutations of the standardised physicochemical matrix preserved its dimension and feature covariance while destroying ligand–score association. An outcome-derived transported rank-7 SVD reconstruction was retained only as a nonpredictive subspace reference, not a capacity ceiling.

For each arm we also correlated the upper triangle of its predicted-component target correlation matrix with the four fixed experimental maps. Equation 7 states the relevant projection algebra. With a shared nuisance basis, independently fitted target coefficients can inherit a low-rank sketch of the training target covariance even when held-out predictive performance is negligible; subsequent column standardisation removes the predicted columns’ scale. Component-map concordance was therefore interpreted only beside the out-of-fold performance metrics and the nuisance ensemble. Chemical-group block reweighting held the fitted out-of-fold surfaces and fixed target panel constant; its ranges are conditional sensitivity intervals, not chemical- or target-population confidence intervals.

2.9 Within-Vina fixed-pose term attribution

To describe which AutoDock Vina terms carry covariance in a fixed-pose subset, we rescored a fixed pose block term by term. A simple random subset of 512 ligands (seed 20260802) from the frozen 15,000-ligand DOCKSTRING support was taken across the nine targets whose published pose archives were available, giving 4,608 cells. Published top poses were converted to PDBQT with Open Babel 3.1.0 using Gasteiger charges and explicit hydrogens, then rescored against the released receptor files. The roundtrip reproduced the released scores for 99.6% of cells within 0.10 kcal/mol at cell-level Pearson 0.9996. We then computed the correlation spectrum, PR, PC1 fraction and mean squared target correlation for each individual Vina term, before and after within-ligand centring, and attributed each term’s share of the shared row-axis covariance. Original docked torsion trees were not published, so this is an attribution of retained poses rather than a causal re-docking ablation.

2.10 Target-pair geometry and the factorial estimand comparison

For matrix X , target-pair geometry was

$$G(X) = \text{corr}_{\text{ligands}}(X), \quad (4)$$

and $G_R(X) = G(\Gamma)$. Agreement between two geometries was Spearman correlation between their $P(P-1)/2$ upper-triangular edges. This statistic describes co-response across the specified ligand supports; it is not a compound-level affinity correlation.

For every experimental panel we computed the four concordances

$$\rho\{G(X), G(E)\}, \quad \rho\{G(X), G_R(E)\}, \quad \rho\{G_R(X), G(E)\}, \quad \rho\{G_R(X), G_R(E)\}. \quad (5)$$

The docking-side increment against the fixed raw endpoint was the third minus the first, and the increment against the fixed centred endpoint was the fourth minus the second. For each fixed endpoint, a paired QAP jointly relabelled the targets of raw and centred docking geometries, held the experimental geometry fixed, and recomputed the difference for 49,999 permutations [44,45].

A separate transformation-matched null used one score-blind simple-random sample without replacement of 15,000 ligands from the 259,579-ligand, experimental-structure-excluded DOCKSTRING reference. Within each of 2,000 draws, ligand identities were independently permuted inside every docking target column; raw and within-ligand-centred docking maps were then derived from the identical permuted matrix while the selected raw or centred experimental geometry remained fixed. This preserves docking target marginals, variances, ties and clipped zeros while destroying ligand-wise target coordination. Each probability compared the observed increment calculated on that same fixed chemical support with its null increments; the full-reference increment is reported descriptively and was not compared directly with the subsampled null distribution.

The sequential experimental-first fraction divided $\rho\{G(X), G_R(E)\} - \rho\{G(X), G(E)\}$ by the joint raw/raw-to-centred/centred change. Because this quantity is order dependent, we also reported the reverse path, the two-factor interaction on the Spearman scale and Shapley-style path-averaged marginal contributions. The latter average each transformation’s marginal contribution across both orders and are descriptive rather than causal.

2.11 Cross-panel experimental geometry and transformation-matched nulls

Cross-panel agreement was the mean of six pairwise Spearman correlations among the four experimental geometries. A global network QAP fixed DAVIS labels and independently permuted complete target labels for PKIS2, PKIS1 and KiRHub, applying identical label permutations to raw and centred geometries. In a separate transformation-matched null, ligand identities were independently permuted within every target column of every panel; raw and centred maps were then recomputed from the same permuted matrices. This preserves target marginals, variances, ties and ceiling patterns while destroying ligand-wise target coordination. Two thousand matched draws evaluated whether the observed centred-minus-raw change exceeded the mechanical change induced by the projection.

The fixed target-pair endpoint comprised pairs in the centred upper 10% in at least two of DAVIS, PKIS2 and PKIS1, giving 15 positives among 190 pairs. KiRHub was not used to define the labels. AUROC and average precision were calculated for KiRHub, sequence, pocket and docking predictors; target-label QAPs permuted complete target identities rather than treating 190 dyads as independent observations.

2.12 Protein-sequence and KLIFS controls

Receptor-domain sequences were extracted from the exact DOCKSTRING PDBQT receptor files and globally aligned with BLOSUM62 to obtain pairwise sequence identity. KLIFS pocket identity was calculated from the aligned 85-residue human kinase-pocket sequences in a checksum-anchored KLIFS snapshot [37, 38]. Partial Spearman statistics rank-transformed the docking, experimental, sequence and pocket edge vectors and residualized predictor and endpoint on sequence plus pocket ranks before correlating the residuals. Unrestricted QAP used 50,000 complete target-label permutations and refit the partial statistic for each draw.

The strict sensitivity restricted label exchanges to KLIFS groups. The panel contained 11 TK targets, three AGC targets, three CMGC targets and one target each in STE, CAMK and an other group. One hundred thousand Monte Carlo permutations were drawn from the 1,437,004,800 possible within-group relabellings. Singleton targets remain fixed under this scheme. The restriction is a conditional sensitivity, not proof that targets are exchangeable within each group.

2.13 Chemical support, centering panel and map recovery

Chemical-support analysis formed lower- and upper-molecular-weight groups at the empirical 25th- and 75th-percentile thresholds, retaining every boundary tie, and compared the resulting centred target geometries. One thousand random disjoint splits of the same sizes formed a matched null. Murcko and Butina cluster bootstraps were reported separately. Butina clustering used RDKit Morgan radius-2, 2,048-bit fingerprints and a Tanimoto similarity threshold of 0.65; an entire cluster and all of its ligands formed the resampling unit [47–50].

An exploratory docking-domain analysis applied the same inclusive lower/upper-threshold split to the full Docking-44 support and a fixed, score-blind 15,000-ligand DOCKSTRING support and computed Spearman agreement between the strict upper triangles of the two residual target-correlation maps. Paired cluster bootstrap resampled the union of chemical groups 300 times. Five hundred simple-random disjoint support pairs calibrated finite-sample agreement. A separate 200-repeat control assigned complete chemical groups to two halves within deciles of group-median molecular weight, greedily balancing row counts; this produced chemically disjoint supports with closely matched molecular-weight distributions. To distinguish genuine between-band change from instability caused merely by restricting chemical support, we also split the low- and high-MW bands separately. Within each band, 500 equal-size row-random disjoint splits and 200 MW-stratified whole-chemical-group-disjoint splits were evaluated on both the raw and row-centred surfaces. Their central 95% repeated-split ranges describe composition sensitivity, not population confidence intervals. We repeated the low/high comparison after the group-held-out seven-descriptor decomposition and across six independently seeded DOCKSTRING supports. A post hoc descriptor- family extension repeated this quantile-threshold split, with boundary ties retained, for heavy-atom count, molecular weight, Labute ASA, TPSA, cLogP, rotatable-bond count and ring count. Because several of these descriptors are correlated, it characterizes the specificity of the observed support shift rather than testing seven independent mechanisms. These comparisons test transfer between observed molecular-size-defined domains, not a molecular-weight-specific causal effect.

The centering-panel sensitivity used the identical 259,579-by-58 DOCKSTRING reference. Under the local contract, the 20 kinase columns were extracted and each ligand mean was calculated over those columns. Under the global contract, the row mean was calculated over all 58 targets before the same 20 columns were extracted. Both were compared with the same centred experimental matrices using 49,999 paired target-label permutations.

For sampling recovery, ligand subsets of 50, 100, 200, 500, 1,000, 2,000 and 5,000 were drawn without replacement from each large matrix. After within-ligand centering, the subset target map was compared with the full source-library map. Each size used 100 draws. Oracle Approximating Shrinkage was a secondary small-sample estimator [51]. A chemical-group-held-out analysis formed a map from the sampled ligands and compared it with a map calculated after excluding their Murcko or acyclic-connectivity groups. Recovery ranges describe repeated samples from the observed library, not a target population or a shifted chemical distribution.

2.14 Mixed-family censored sensitivity

The primary Secondary Pharmacology Database analysis used 12 mapped target groups and 59 of their 66 possible target pairs; alternative exact-only and common-rectangle estimands used smaller supports [36]. Released IC₅₀ bounds were analyzed as percentile-ranked bounds and as censor-aware binary outcomes at 10 and 30 μ M. Pairwise-complete experimental correlations retained censoring metadata. Unrestricted and family-preserving target-label permutations were evaluated separately. The source subset is 91.6% right-censored, so this analysis was not combined with the dense kinase panels.

2.15 Dense and sparse observed-pair sensitivities

For dense operational checks, full Standard InChIKeys were recomputed with the pinned RDKit version from the released DOCKSTRING, DAVIS and PKIS2 SMILES. Duplicate raw DOCKSTRING profiles and duplicate experimental identities were collapsed cell-wise by the median. The 21 shared kinase targets were the fixed 20-target geometry panel plus PTK2/FAK. All DOCKSTRING rows whose recomputed 14-character Standard-InChI connectivity block occurred in the corresponding evaluation panel were removed before fitting target means, target standard deviations, the grand mean or residual target standard deviations. Positive reference and evaluation Vina scores were clipped to zero under the primary contract.

The DAVIS primary estimand was the equal-ligand mean of within-ligand pairwise concordance on the dense 59-by-21 exact-key block. Lower Vina scores predicted higher pK_d; exact pK_d ties and pairs for which both measurements were at the released pK_d-5 floor were excluded. The PKIS2 estimand used the dense 154-by-21 block and retained experimental target differences greater than 10 percentage points; higher single-dose inhibition/displacement and lower Vina score indicated preference. Predicted score ties received half credit. These are observed-target

ordering checks, not retrieval of unmeasured targets; PKIS2 is not an affinity measurement.

Representations were absolute Vina, target-centred unscaled, column-standardized, target-centred divided by the reference residual target SD, and the scaled two-way residual. Every evaluation-ligand row mean used by the dense scaled residual rule was calculated over the same 21 shared targets; all target offsets and scales came from the leakage-free reference. The ligand-invariant docking target prior assigned every evaluation ligand the target means fitted on the same leakage-free DOCKSTRING reference. Paired contrasts included absolute Vina minus this prior and the full path from absolute scores through target-offset removal, residual-SD scaling and ligand-row correction. For the last two representations, Eq. 9 follows directly from the score definitions. Uncertainty used 5,000 ligand, Murcko/connectivity-cluster and Butina- cluster bootstraps with seed 0. Acyclic Murcko groups were identified by recomputed Standard-InChI connectivity block; Butina used radius-2, 2,048-bit Morgan fingerprints and Tanimoto similarity 0.65. Delete-one-target analyses refitted every offset, row mean and scale on the remaining 20-target reference; their normal intervals are descriptive composition sensitivities because the 21 targets are fixed and non-random.

DAVIS outcome-conditioned sensitivities separated pairs with two uncensored pKd values from floor-versus-uncensored pairs. The primary PKIS2 sensitivity required both target values to be at least 65% inhibition/displacement and retained the aggregate margin of more than 10 percentage points; a secondary sensitivity relaxed only that margin to exact non-ties. These strata were selected after examining the aggregate results, and the transformation-path and stratum intervals were not adjusted for multiplicity. They are reported as mechanistic and hypothesis-generating decompositions, not alternative primary endpoints.

The original Docking-44-ChEMBL benchmark was retained only as a sparse support sensitivity and was extracted from ChEMBL release 34 [39,40]. The primary block retained exact standard relations, aggregated repeated compound-target measurements by the median and included Ki, Kd, IC50 and EC50. Only experimentally observed cells were evaluated; no missing activity was imputed. For each ligand and unordered pair of observed targets, a prediction was correct when the more negative docking score corresponded to the larger pChEMBL value. Experimental ties were excluded, predicted ties received half credit, and per-ligand pair accuracy was averaged so that ligands received equal weight.

A pair-weighted sensitivity instead gave every non-tied comparison equal weight. The ligand-identity null permuted complete docking-score rows relative to the experimental profiles. In 25

chemical-de-redundancy repeats, one ligand was sampled from each Murcko cluster before 200 such row permutations; the resulting probability distribution is a sensitivity across representative supports rather than a single cluster-level test.

For this sparse sensitivity only, all docking transformation parameters were estimated from the 12,514 Docking-44 ligands outside the 137-ligand evaluation cohort. Representations included absolute Vina, target-centred unscaled, two-way-centred unscaled, column-standardized and residual- standardized scores. The last representation subtracts row and column effects on the score scale and divides each target by its externally estimated residual standard deviation. Its ligand row mean was calculated over all 44 Docking-44 targets before the 38 targets observed in the ChEMBL benchmark were extracted; this 44-target centering panel is part of the operational score definition. Equation 8 was checked numerically by comparing target-centred and two-way-centred unscaled scores.

Target-only controls separated ligand-specific ranking from target marginals. The docking prior repeated each target's mean clipped Vina score from the 12,514-ligand external reference for every evaluation ligand. An external experimental prior analogously used targetwise mean pChEMBL from 96,744 ChEMBL ligands after excluding the evaluation cohort. The cohort outcome prior was cross-fitted within the 137-ligand benchmark: for each ligand, targetwise means were recomputed after excluding its complete Bemis–Murcko scaffold cluster, with absent targets falling back to the external experimental prior. This final control is target-only and scaffold-held-out but outcome-informed and cohort-specific; it is not an externally deployable predictor. Coverage strata evaluated the already cross-fitted predictions without refitting them.

A one-factor centering-panel sensitivity recomputed each evaluation ligand's row mean over the 38 evaluated targets. Per-target missing-score means, target offsets, the training grand mean and residual standard deviations were held at their original estimates from the 12,514-ligand, 44-target reference. This isolates dependence on the six row-mean-only context targets; it is not a newly fitted 38-target normalization. Paired ligand, Murcko and Butina bootstraps used the same 5,000-replicate contract as the primary comparison.

Sparse-sensitivity uncertainty used 5,000 ligand bootstraps, 5,000 Murcko-cluster bootstraps and 5,000 Butina-cluster bootstraps; conservative intervals are unions of the two chemical-cluster intervals. Leave-one-target-out values describe composition sensitivity conditional on the observed 38-target panel. Human binding K_i/K_d , same-endpoint, minimum activity-difference, minimum target-coverage and measurement-discordance analyses were secondary. Equivalence

readings of these intervals are post hoc throughout: no margin was preregistered, and every margin reported here was chosen after the corresponding benchmark had been computed. We therefore report a margin sensitivity curve rather than a single threshold, and treat the paired difference and its interval as the primary result.

2.16 Software, numerical provenance and reproducibility

Analyses used Python 3.12.11, NumPy 1.26.4, pandas 2.3.2, SciPy 1.16.2, scikit-learn 1.7.2, Matplotlib 3.10.6 and RDKit 2025.03.6. Random seeds are recorded per analysis rather than represented by one global seed. The machine-readable manuscript ledger contains each headline value, analysis contract, source path and SHA-256 checksum. `analysis/build_manuscript_evidence.py` constructs the ledger, `analysis/make_reported_results.py` generates high-risk LaTeX numbers, and `analysis/make_manuscript_figures.py` generates the five main figures. The generic `analysis/spectral_audit.py` command-line program applies the spectral portion of the audit to a dense CSV or TSV matrix.

The default report-layer build verifies the frozen matrices and result artifacts, then regenerates the manuscript ledger, tables, figures and PDFs from the checksum-verified artifacts. It does not silently rerun source-restricted analyses or the large upstream docking campaigns. Dataset-specific scripts retain the statistical contracts and can be rerun from redistributed inputs or checksum-identified source workbooks under their respective terms. Input and result manifests, a pinned environment, Dockerfile, Makefile and tests are provided with the code and derived data.

OpenAI Codex and Anthropic Claude were used during manuscript preparation for code review, literature-search assistance and language editing. They were not treated as sources of scientific evidence. The authors inspected the underlying data and code, verified the reported claims and citations, and retain responsibility for the work.

3 Results

3.1 A shared ligand axis dominates both Vina matrices

The column-correlation spectrum of Docking-44 was strongly concentrated. Its first principal component (PC1) explained 73.3% of variance and its PR effective dimension was 1.834 (Fig. 1a). DOCKSTRING-58 gave the same qualitative pattern at larger scale: PC1 explained 65.9% and PR dimension was 2.266 (Fig. 1b). The PC1 loading vectors were close to a uniform target

vector (cosines 0.970 and 0.969); 43 of 44 and 57 of 58 loadings had the same sign. PC1 scores correlated 0.998 and 0.997 with the mean of the column-standardized scores for each ligand. The dominant mode was therefore a shared ligand-wide score axis rather than one target or protein family.

For a P -target correlation matrix C with eigenvalues λ_k , PR has the exact form

$$r_{\text{PR}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}, \quad (6)$$

where the bar is over unique off-diagonal target pairs. PR is thus a monotone re-expression of mean squared inter-target correlation on an effective-component scale; it is not an independent statistic, an algebraic rank or a docking-quality score.

Removing the mean score of each ligand increased PR to 9.303 and 18.206 (Fig. 1a–c), and turned the predominantly positive correlations of the uncentred matrices into roughly balanced positive and negative ones. The residual maps nevertheless remain strongly dependent, and the useful way to say how strongly is as an effect size rather than a test. Randomizing each ligand’s residual direction within the target-zero-sum subspace, while preserving its residual norm, gives PR near the algebraic ceilings of 43 and 57 (medians 42.628 and 56.612 on fixed 12,000-ligand supports). The observed residual PR values of 9.295 and 18.158 therefore represent a compression of roughly 43 effectively independent target directions into 9, and of 57 into 18. Because that null removes aligned target structure apart from the zero-sum and row-norm constraints, its rejection at $N = 12,000$ is a foregone conclusion; the informative quantity is the compression ratio, not the probability. Supplementary Information reports the full battery of nulls.

The direction, but not the magnitude, of the PR increase after centring was robust to choosing a correlation or covariance spectrum. Covariance-spectrum PR increased from 1.915 to 6.911 in Docking-44 and from 2.474 to 8.235 in DOCKSTRING-58; the lower covariance residual values show that the magnitude depends on whether target variances are standardized. The direction was also robust to missing-data handling and clipping: Docking-44 residual correlation PR was 9.303, 9.528 and 9.663 under target-mean imputation, target-median imputation and observed-cell additive least squares; and retaining the rare positive DOCKSTRING scores instead of clipping them moved residual PR from 18.206 only to 18.362. Complete cases gave 13.970, but the 9,297 complete ligands are a chemically selected subset. Full missingness, covariance,

clipping, target-subsampling and chemical-cluster sensitivities are in Supplementary Information.

Broad protein-family labels did not resolve the remaining target dependence: the within-minus-between-family differences in mean absolute residual correlation were 0.030 and 0.017, with label-permutation $p = 0.073$ and 0.067 for Docking-44 and DOCKSTRING-58, respectively (Supplementary Fig. S3). The residual map is therefore correlated, but is not demonstrated here to encode broad family organization.

What survives centring has a descriptive ligand-property association. Per-target regressions of Vina score on molecular weight and RDKit rotatable-bond count gave median conditional molecular-weight slopes of -0.350 and -0.778 kcal/mol per 100 Da and median rotatable-bond slopes of +0.100 and +0.153 kcal/mol per bond, the flexibility slope being positive for 44/44 and 57/58 targets. Target-wise molecular-weight residual slopes aligned with residual-PC1 loadings (cosines 0.963 and 0.929). Row centring therefore removes the average ligand-size tendency but keeps the target-to-target heterogeneity in how each score responds to size and flexibility. That heterogeneity is compatible with scoring bias, pocket-dependent physicochemical preference, or both; this analysis does not distinguish them. These regressions are descriptive: RDKit rotatable-bond count is not Vina's internal torsional term. Docking-44 used the full mean-imputed support, with the alignment persisting after excluding 4mq5, the target with the most missing scores; DOCKSTRING used one fixed, score-blind 15,000-ligand descriptor support (Supplementary Fig. S5).

Five-fold chemical-group-held-out fits confirmed that low-dimensional ligand features have non-zero held-out predictive performance for the residual score surface: removing seven ordinary descriptors reduced mean squared target correlation by 46.9% and 53.6%. These non-additive operational ratios are not shares of correlation explained. The rank-matched controls below show that the predictive basis is not specific to those descriptors and that correlation geometry of a fitted component has a separate scale-free nuisance-projection failure. Ligand-property controls serve the same bounding role in structure-based benchmark design [10–13].

A secondary fixed-pose sensitivity is reported in Supplementary Information. On 2,000 ligands and 9 targets, five of six non-Vina scorers met the stated shared-axis criteria, whereas PLECScore-linear falsified a universal-threshold claim; all functions were evaluated on Vina-selected poses. A signed within-Vina attribution on 512 ligands found that gauss2 contributed 0.715 of the shared-axis covariance and 0.525 of the residual Frobenius inner product, but deletion was evaluated arithmetically on fixed poses. Neither analysis is an independent redocking comparison, a

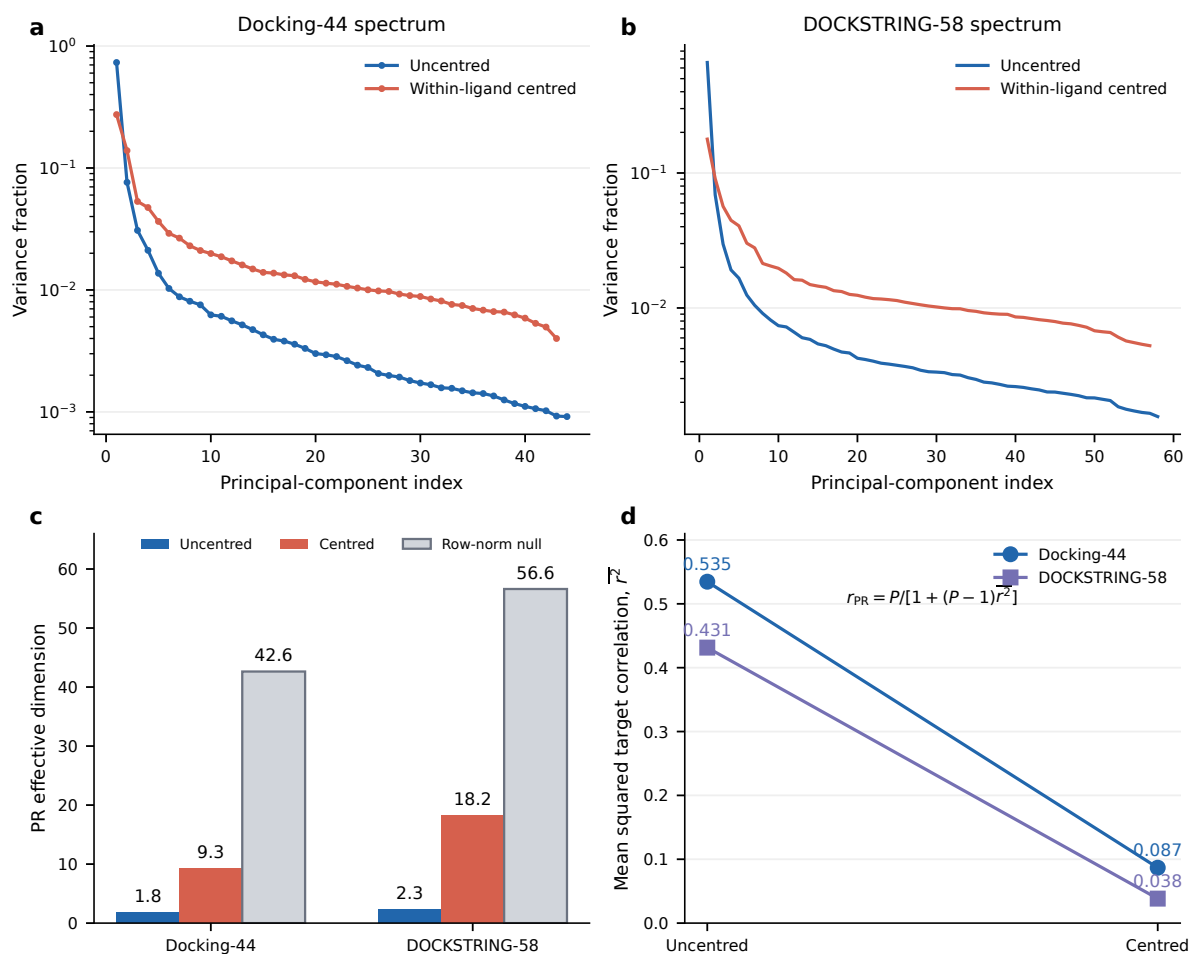


Figure 1: A shared ligand axis concentrates both Vina score matrices. **a,b**, Normalized eigenvalue spectra before and after within-ligand centering. The numerical zero mode introduced by centering is omitted from the logarithmic plot. **c**, PR effective dimensions and row-norm-preserving null medians on matched 12,000-ligand supports. The residual algebraic ceilings are 43 and 57. **d**, Mean squared target correlations before and after centering. Equation 6 maps each value exactly to PR; the full spectra retain information about how dependence is distributed among modes.

non-Vina replication of pose selection or a causal term-ablation experiment.

The two score matrices were generated in separate projects and computational pipelines, not sampled as statistically independent observations. Two authors of the present study also contributed to the Docking-44 source study. They shared 582 full Standard InChIKeys, 804 connectivity blocks and 1340 non-empty Bemis–Murcko scaffolds. Their target lists contained 9 mapped protein identities, of which 3 used the same receptor structure. The replicated spectral pattern should therefore be read as agreement across two different data-generation pipelines with quantified overlap.

3.2 Exploratory molecular-size splits alter the map, which is recoverable within each source library

Target-pair geometry is a property of a target panel evaluated over a specified chemical library. In exploratory, post hoc analyses within the observed source libraries, this dependence was pronounced on the full Docking-44 support and a fixed 15,000-ligand DOCKSTRING support. Residual maps from the lowest and highest molecular-weight quartiles correlated only 0.205 (chemical-group bootstrap 0.185–0.219) in Docking-44 and 0.351 (0.326–0.363) in DOCKSTRING-58, compared with 0.991 and 0.994 for chemically disjoint controls with closely matched molecular-weight distributions (Fig. 2a). The same between-band shift was already present before row centring: raw-map agreements were 0.227 (0.209–0.246) and 0.409 (0.384–0.432), versus 0.995 and 0.998 in their MW-matched controls. Row centring therefore did not manufacture the domain shift.

Restriction-matched controls excluded a second mechanical explanation. When each extreme MW band was split internally while keeping entire chemical groups disjoint, residual maps agreed at 0.980 and 0.978 in the low and high Docking-44 bands, and at 0.978 and 0.966 in DOCKSTRING-58; even the lowest endpoint of the four repeated-split ranges was 0.957. Corresponding raw-map means were 0.989/0.989 and 0.991/0.979. Thus correlation maps were reproducible within either restricted band while differing sharply between bands. Target-pair signs changed for 42.5% and 36.7% of edges. Removing the seven-feature component in strictly group-held-out folds increased cross-domain agreement to 0.489 and 0.437, respectively, but did not approach the matched-domain controls. Size-matched random disjoint supports agreed at 0.991 and 0.990, respectively. Across six independently seeded DOCKSTRING supports the unadjusted cross-domain value remained between 0.325 and 0.361. Across post hoc extreme splits of the seven inspected descriptors, the three strongest shifts in both matrices formed a correlated molecular-size family: Labute ASA, molecular weight and heavy-atom count gave residual-map agreements of 0.200–0.207 in Docking-44 and 0.322–0.361 in DOCKSTRING-58. The remaining descriptor splits were weaker (0.703–0.887 and 0.787–0.831, respectively), although all lay below the same dataset-level MW-matched composition reference rather than a descriptor-matched null. Residual PR itself moved in opposite directions across the MW split: 13.625 to 10.147 in Docking-44, but 19.552 to 23.800 in DOCKSTRING-58. These post hoc analyses distinguish molecular-size-domain drift from finite sample size or generic chemical-group disjointness; molecular weight defines the tested domains but is not established as the sole

cause, and the PR reversal precludes a monotone rank-quality interpretation.

Experimental maps showed the same qualitative boundary. Splitting compounds by molecular-weight quartile changed the map more than matched random disjoint splits in PKIS2 and PKIS1: low- versus high-MW geometry agreement was 0.333 and 0.547, compared with random-split means of 0.845 and 0.833 ($p = 0.001$ for each; Fig. 2b). DAVIS, with only 18 compounds in each extreme quartile, gave 0.531 against a random mean of 0.619 ($p = 0.173$). The direction of individual edge changes was not consistent across panels, so molecular weight is a support marker rather than an identified mechanism.

The target set used to compute each ligand mean had a modest effect. Computing row means over the local 20 kinases and over all 58 DOCKSTRING targets before extracting the same 20 produced maps with Spearman correlation 0.973. Concordance with the experimental panels changed by -0.008 to -0.019 , with no resolved positive all-58 gain (paired $p = 0.678$ – 0.925 ; Supplementary Information). The local 20-target definition is primary because it matches the validation panel.

Although conditional, each observed source-library map was closely reproduced by far fewer ligands than the full matrix. Across 100 simple-random replicates of 200 same-distribution ligands, mean agreement with the full map was 0.918 for DOCKSTRING-58 (2.5–97.5% range 0.885–0.938) and 0.942 for Docking-44 (0.898–0.962). At 500 ligands the means were 0.965 and 0.976 (Fig. 2c). When sampled chemical groups were excluded from the reference map, 200-ligand agreement remained 0.916 for DOCKSTRING-58 and 0.927 for Docking-44 (Fig. 2d). These panels require 11,600 rather than 15,083,480 score cells for DOCKSTRING and 8,800 rather than 556,644 for Docking-44, arithmetic reductions of approximately 1,300-fold and 63-fold.

The 2.5–97.5% replicate ranges at 200 ligands were 0.053 wide for DOCKSTRING-58 and 0.064 for Docking-44, comparable to or larger than the 0.001–0.045 docking-side increments in the fixed centred-endpoint factorial. Thus 200 ligands supported coarse reconstruction and support-shift diagnosis here, but not resolution of small between-map or method contrasts.

This is a descriptive, support-specific sampling-reliability result. It shows that the target-pair map of a specified scoring-function, receptor and chemical-library combination can be measured without docking the entire source library. It does not show that the map is biologically correct, that 200 ligands recover individual target profiles, or that the estimate transfers to a shifted chemical library. Nor does it establish 200 ligands or any observed agreement value as a universal

accuracy threshold for a different library, receptor panel or scorer.

3.3 Centred experimental geometries agree across panels, but the increment is mechanical

We defined target-pair geometry as the upper triangle of the target correlation matrix. Experimental matrices were restricted to the same ordered 20 kinases and contained 72 DAVIS compounds, 645 PKIS2 compounds, 360 PKIS1 compounds and 92 KiRHub compounds. The panels differ in compounds, assay technologies and endpoints; they are separately published assay panels rather than independent samples from a target superpopulation. DAVIS is also strongly measurement-limited: 965 of 1,440 cells (67.0%) lie at the 10,000-nM Kd upper bound, corresponding to a pKd floor of 5. A binary above-floor-status geometry correlated $\rho = 0.885$ with its continuous pKd geometry; centred DOCKSTRING remained associated with that binary geometry ($\rho = 0.292$, $p = 0.00426$). Among the three panels with molecular structures, pairwise intersections contained only 6, 1 and 7 Standard-InChI connectivity blocks for DAVIS-PKIS2, DAVIS-PKIS1 and PKIS2-PKIS1, respectively, with 0 shared by all three. KiRHub structures were not available, so its remaining chemical overlap could not be audited at structure or scaffold level.

The four within-ligand-centred experimental geometries agreed strongly. Their mean pairwise Spearman correlation was 0.763, compared with 0.632 for the uncentred geometries (Supplementary Table S10). KiRHub, inspected after a 15-pair endpoint had been fixed from the other three panels, correlated 0.834 with their mean centred geometry. The value was 0.817 after removing the 11 exact-name DAVIS overlaps and 0.802 after additionally adjusting that overlap-excluded comparison for receptor-domain sequence identity. The locked 15-pair endpoint was also recovered by centred KiRHub geometry with AUROC 0.963 and average precision 0.694 (target-label QAP $p < 0.001$ for each). These centred levels support cross-panel stability of the fixed-panel endpoint. Paired target-label relabelling also resolved the centred-minus-raw KiRHub gain (AUROC $p = 0.04928$; AP $p = 0.02306$; Holm-adjusted $p = 0.04928$ and $p = 0.04612$), but that gain did not exceed its transformation-matched mechanical null ($p = 0.748$ and $p = 0.997$). We therefore claim the centred KiRHub level, not an informative gain from centring. The absolute centred KiRHub AUROC and average precision each exceeded their transformation-matched null ($p = 0.00050$ for both).

Centering itself, however, mechanically changes correlations. A global network QAP that

destroyed target correspondence across panels supported both absolute centred agreement and its increment over raw agreement ($p < 0.001$ and $p = 0.0020$). A complementary transformation-matched null independently permuted ligand identities within every target column and then repeated both transformations; it produced a mean centred-minus-raw change larger than the observed one ($p = 0.947$). Its mean raw agreement was -0.0002 , not the observed 0.632 , so this null evaluates an unconditional mechanical centering gain rather than conditioning on the already aligned raw maps. Absolute centred agreement still exceeded it ($p = 0.00050$). The matched raw baselines were near chance, rather than the already high observed raw metrics, so both gain tests are unconditional.

The single number that settles how to read this is the ratio of the observed increment to the mechanical one. Centring raised mean cross-panel agreement by $+0.131$, from 0.632 to 0.763 ; the transformation-matched null, which applies the identical pair of transformations to ligand-permuted columns, produced a mean increment of $+0.179$. The observed gain is thus only 0.73 times what the transformation generates on data with no shared target structure at all, and the level—not the increment—is what carries information here. We therefore claim only the level: the four centred experimental geometries agree strongly and that agreement exceeds its matched null. We do not claim that centring reveals a more faithful experimental geometry. A concrete alternative explanation is available and untested: the panels screen different compound libraries, and centring removes compound promiscuity, which differs between libraries, so the rise in agreement may reflect library heterogeneity rather than biology.

3.4 Docking–experiment agreement is path dependent

For each assay panel, we evaluated all four cells of a 2×2 design: uncentred or within-ligand-centred docking geometry crossed with uncentred or centred experimental geometry (Table 2; Fig. 3a). Raw docking against raw experiment ranged from -0.215 to 0.050 . These negative values are not a score-direction error: multiplying every activity or every docking column by -1 leaves all within-matrix target correlations unchanged. They instead show that the ordering of target-pair correlations differs when ligand main effects are retained.

Against the fixed centred experimental endpoint, raw docking geometry was already positively concordant in all four panels (0.173 – 0.276). Within-ligand centering of docking raised these values to 0.174 – 0.322 . The conditional increments were 0.016 for DAVIS, 0.044 for PKIS2, 0.001 for PKIS1 and 0.045 for KiRHub; none was resolved by a paired target-label QAP ($p = 0.335$,

Table 2: Complete docking-by-experimental estimand decomposition on the same 20 kinases and 190 shared target pairs. Entries are Spearman correlations between those 190 target-pair edges. Each Δ_D compares centred and raw docking against the stated fixed experimental endpoint; each cell gives the full-reference increment followed by the matched-15,000-ligand-support increment. p_L is its one-sided paired target-label QAP probability from 49,999 permutations and p_M is the transformation-matched probability for the second increment from 2,000 independent-column permutations on a fixed 15,000-ligand support. Differences were calculated from unrounded correlations and can therefore differ slightly from subtraction of the displayed entries.

Panel	raw/raw	raw/centred	centred/raw	centred/centred	$\Delta_D^{F/M} E_{raw}$	p_L	p_M	$\Delta_D^{F/M} E_{ctr}$	p_L	p_M
DAVIS	0.050	0.263	0.219	0.280	0.168/0.162	0.01512	0.00300	0.016/0.010	0.335	0.690
PKIS2	-0.215	0.243	0.033	0.287	0.249/0.243	0.00152	0.00050	0.044/0.034	0.146	0.172
PKIS1	-0.159	0.173	0.059	0.174	0.218/0.215	0.00536	0.00050	0.001/-0.006	0.462	0.513
KiRHub	-0.029	0.276	0.235	0.322	0.264/0.261	0.00008	0.00050	0.045/0.041	0.126	0.073

0.146, 0.462 and 0.126; Fig. 3a). All four increments also failed to exceed the transformation-matched mechanical null ($p = 0.690, 0.172, 0.513$ and 0.073). All four full-support increments were positive, so the result does not establish a zero docking-side effect. The matched null began at near-zero docking–experiment concordance rather than the observed 0.173–0.276; it therefore tests an unconditional mechanical centering gain, not a gain conditioned on the already aligned raw maps. The panels share the same 190 target pairs and their probabilities were not pooled. The 95% target-label null intervals all spanned zero, with an overall envelope from -0.086 to 0.104 ; these are null intervals rather than confidence intervals for a target population.

Against the fixed uncentred experimental endpoint, the same docking transformation gave larger increments: 0.168 for DAVIS, 0.249 for PKIS2, 0.218 for PKIS1 and 0.264 for KiRHub. All four were resolved by the corresponding paired target-label QAP ($p = 0.01512, 0.00152, 0.00536$ and 0.00008) and exceeded the transformation-matched null on its fixed 15,000-ligand support ($p = 0.00300, 0.00050, 0.00050$, and 0.00050). All four target-label probabilities remained below 0.05 after Holm correction across these four raw-endpoint tests. Thus the apparent magnitude and statistical resolution of the docking transformation depend on which experimental geometry is held fixed.

The joint raw/raw-to-centred/centred comparison changes predictor and endpoint together. Along the experimental-first path, 87.0–99.7% of that change occurred before the docking predictor was centred. This percentage is order dependent. Reversing the order gave larger docking-first increments, and the two-factor interactions on the Spearman scale were negative (-0.152 to -0.219 ; Supplementary Table S7). A path-averaged descriptive allocation assigned 55.8–70.9% of the total change to the experimental estimand (Supplementary Table S8). These calculations are descriptive, not causal, and their numerical allocation depends on the chosen

concordance scale; applying a nonlinear transform such as Fisher’s z would allocate the same four cells differently. The fixed-endpoint comparison in the preceding paragraphs isolates the docking transformation on each endpoint. Their contrast is the estimand dependence: the centred experimental geometry is the target-pair co-response after removal of ligand main effects, whereas the uncentred geometry retains them.

For DAVIS, PKIS2 and PKIS1, direct reuse of experimental ligands did not explain the fixed-panel association. We removed the union of 1,050 experimental Standard-InChI connectivity blocks from the DOCKSTRING reference, excluding 481 of 260,060 rows. On the same strict common-20 estimand, excluding a further 3,039 rows with matching cyclic Murcko scaffolds changed centred geometry concordance by at most 0.0017, and five score-blind 15,000-ligand reference samples closely reproduced the full-reference values. KiRHub reports names but not molecular structures, so an equivalent identity- or scaffold-based exclusion was unavailable for that panel (Supplementary Information).

3.5 Predictive feature structure does not validate a component correlation map

Seven ordinary ligand descriptors predicted nontrivial score structure out of chemical group. Across repeated group partitions, their removal reduced mean squared residual target correlation by 0.462–0.470 in Docking-44 and 0.527–0.546 in DOCKSTRING-58; across five score-blind DOCKSTRING supports the support means ranged from 0.528 to 0.545. These are operational before/after ratios, not additive fractions of correlation explained. Each target has its own fitted coefficient vector, so the result establishes a low-dimensional target-indexed ligand-feature response rather than a receptor-independent physicochemical mechanism.

Capacity-matched controls on the common-20 support after excluding structural overlaps with DAVIS, PKIS2 and PKIS1 made that boundary sharper. The seven physicochemical descriptors had mean out-of-fold target $R^2 = 0.107$ and reduced mean squared residual target correlation by 0.234. A locked seven-dimensional random projection of count-Morgan features gave $R^2 = 0.090$ and reduction 0.238; reductions across 20 fixed projection seeds ranged from 0.176 to 0.260, and 3/20 exceeded the physicochemical value. Thus the predictive signal is not specific to the named descriptors. It can be represented by a generic low-dimensional chemical-feature basis that retains molecular size and local topology.

A more important negative control exposed a failure mode in validating fitted components by their target-correlation maps. Let H be a shared q -dimensional ligand basis and Y the residual

ligand-by-target score surface. Target-wise least squares in one training fold gives

$$\hat{B} = (H_{\text{tr}}^T H_{\text{tr}})^{-1} H_{\text{tr}}^T Y_{\text{tr}}, \quad \hat{Y}_{\text{te}} = H_{\text{te}} \hat{B}. \quad (7)$$

Even when H is independent of ligand chemistry and scores, the target-specific columns of $\hat{B}$ form a low-rank random sketch of the training score covariance. Out-of-fold predictive performance then tends towards zero, but column correlation divides away the predicted columns' scale. In our stable seven-dimensional ligand-identity hash control, mean out-of-fold target R^2 was -0.00004 and mean-squared-correlation reduction was 0.00002, yet the fitted component map correlated 0.103–0.348 with the four experimental maps, matching or exceeding the physicochemical component in three of four panels (physicochemical range 0.097–0.288). Across 20 row permutations of the same seven-dimensional physicochemical matrix, mean target R^2 remained -0.00005–0.00002 and correlation reduction remained -0.00007–0.00007, while component- map agreement ranged from -0.072 to 0.564 across panels and seeds (Fig. 4; Supplementary Information). These are fixed-seed algorithmic ranges, not confidence intervals.

The feature fits therefore support two distinct conclusions. Physicochemical and count-Morgan bases genuinely predict part of the docking score surface out of group, whereas the external concordance of a fitted component's correlation map is not, by itself, evidence that the component is predictive, chemical or biological. Such a map must be accompanied by an out-of-fold predictive-performance metric and a nuisance-basis control. We consequently do not assign the full docking–experiment concordance to the descriptor component, its correlated remainder, or a mediated fraction.

3.6 Structural and measurement controls bound the target-pair map

DAVIS required a separate measurement-limit check. Excluding the four targets with more than 90% of values at the reporting floor (AKT1, AKT2, MAPK1 and MAPKAPK2) left 16 targets. On that support centred docking geometry correlated 0.210 with continuous DAVIS geometry ($p = 0.030$ under 49,999 target-label permutations) and 0.197 with binary above-floor status geometry ($p = 0.037$). The two experimental geometries correlated 0.811 (Fig. 3d), indicating that the continuous DAVIS map largely tracked whether measurements lay above the reporting floor on this support. This sensitivity supports a coarse co-response or reporting-status pattern on the fixed panel; it does not turn floor-censored DAVIS values into finely resolved quantitative

818 affinities.

819 Sequence and pocket similarity produced comparable or higher point estimates than Vina
820 geometry for the experimental target-pair pattern (Fig. 3b). A target pair was labelled positive
821 when it lay in the centred upper decile in at least two of DAVIS, PKIS2 and PKIS1; this yielded 15
822 of 190 pairs and was fixed before KiRHub was inspected. Receptor-domain sequence identity
823 retrieved those pairs with AUROC 0.718 and average precision 0.426; KLIFS 85-residue pocket
824 identity gave 0.708 and 0.486. Raw Vina geometry gave 0.597 and 0.236, while centred Vina gave
825 0.710 and 0.301. Because the positives were defined from centred experimental geometries, this
826 raw-versus-centred retrieval gap is conditional on the chosen endpoint and cannot be interpreted
827 as a causal or transformation-neutral benefit of centring. In the declared 5–25% threshold-
828 family omnibus, the mean centred-minus-raw gain was unresolved (target-label QAP $p = 0.061$
829 for AUROC and $p = 0.076$ for average precision), limiting the result to an upper-tail finding
830 rather than a threshold-general improvement. With only 15 positives, these values are bounding
831 diagnostics rather than evidence that one predictor outperforms another. Binary indicators of
832 shared KLIFS group or family also retrieved this endpoint above their unrestricted label nulls
833 (Supplementary Information), reinforcing the need to condition biological interpretations on
834 kinase taxonomy.

835 Continuous adjustment gave the same boundary. After rank-linear adjustment for both
836 receptor-domain sequence identity and KLIFS pocket identity, centred Vina geometry retained a
837 partial Spearman correlation of 0.225 with the mean of DAVIS, PKIS2 and PKIS1 (unrestricted
838 target-label QAP $p = 0.014$) and 0.280 with KiRHub ($p = 0.0068$). When target labels could be
839 exchanged only within KLIFS kinase groups, the corresponding probabilities were 0.197 and
840 0.166 (Fig. 3c). The association therefore remains positive after the tested continuous similarities
841 under an unrestricted null, but it is unresolved within coarse kinase groups. Group-level kinase
842 organization remains a plausible explanation; the analysis does not establish independent within-
843 family discrimination. The effective randomization units are targets, not 190 independent dyads:
844 the restricted null moves 17 labels in blocks of 11, 3 and 3 and leaves three singleton targets
845 fixed. Its empirical one-sided 5% critical correlations after sequence and pocket adjustment were
846 0.360 for the three-panel mean and 0.376 for KiRHub; the corresponding null medians were
847 0.103 and 0.161. Crossing the critical boundary would require increases of 0.136 and 0.096 over
848 the observed values. These are fixed-matrix randomization-resolution diagnostics, not power
849 calculations or population confidence limits.

A 12-target mixed-family sensitivity was 91.6% right-censored. Family-preserving QAP did not resolve centred-minus-row gains for the released-bound, 10- μ M or 30- μ M endpoints (one-sided $p = 0.729, 0.939$ and 0.714 , respectively). The corresponding point estimates and unrestricted tests are reported in Supplementary Information.

3.7 Target-pair geometry is not a ligand-wise target-ranking rule

Within-ligand centering has an exact operational limitation. If $R_{ij} = X_{ij} - \bar{X}_{i\cdot}$, then for any two targets j and k ,

$$R_{ij} - R_{ik} = X_{ij} - X_{ik}. \quad (8)$$

Subtracting one ligand-specific constant cannot change that ligand’s target order. Relative to absolute scores, target-specific offsets and scales can change order; the row subtraction alone cannot. Accordingly, target-centred and two-way-centred unscaled scores had exactly the same broad-benchmark observed-pair concordance of 0.533.

The primary operational test used the broad DOCKSTRING–ChEMBL overlap. It retained 6,557 ligands with at least two exact-relation experimental cells; 6,480 contributed at least one non-tied comparison and 77 did not. The evaluated support contained 31 of the 58 DOCKSTRING targets, 18,020 observed cells and 25,214 within-ligand target-pair comparisons. It measures concordance only on targets that ChEMBL happened to report for each ligand, not retrieval of unmeasured targets.

Absolute Vina, column-standardized scores and scaled two-way residuals gave mean observed-pair concordances 0.537, 0.530 and 0.529 (Fig. 5a). Thus absolute Vina had the highest aggregate point estimate among the tested docking-derived representations, but not among all references. A target-only prior fitted after holding out the evaluated ligand’s complete scaffold cluster reached 0.604, compared with 0.524 for a prior fitted on the external docking library and 0.480 for a cohort-excluded ChEMBL prior. The scaffold-held-out cohort prior uses outcomes from other benchmark ligands and is a diagnostic of target prevalence and assay-selection structure, not a deployable ligand-specific predictor. Absolute Vina exceeded the ligand-invariant target-only prior fitted on the external DOCKSTRING reference by only 0.013 in paired comparisons, with a conservative chemical-cluster 95% interval from -0.011 to 0.038. The broad benchmark therefore did not resolve ligand-specific ordering information beyond this target-only docking baseline.

The paired residual-minus-absolute difference was -0.008. Chemical-cluster resampling,

conditional on the 31 fixed target labels, gave a 95% interval of -0.022 to 0.006. Delete-one-target jackknifing instead probed panel composition and gave the much wider approximation -0.056 to 0.030 (Fig. 5b). These are deliberately separate uncertainty layers: neither a target-label QAP nor the transformation-matched geometry null is a sampling interval for ligand-wise concordance. The corresponding residual-minus-column-standardized difference was near zero and unresolved under both layers (Fig. 5b,c).

The less assay-heterogeneous, human-binding Ki/Kd-restricted arm retained 1,056 ligands, of which 1,049 contributed 3,955 non-tied pairs over 31 targets. Absolute and scaled-residual concordance were 0.559 and 0.520, a paired difference of -0.040 (chemical-cluster 95% interval -0.091 to 0.003; target-jackknife approximation -0.133 to 0.029; Fig. 5b). This arm can still compare a Ki-derived cell with a Kd-derived cell within one ligand and is therefore endpoint-restricted, not homogeneous. Of 3,955 pairs, 3,610 (91.3%) compared the same single endpoint, 33 (0.8%) were unambiguous Ki-Kd comparisons and 312 (7.9%) involved at least one cell pooled across both endpoint types. The point estimate favoured absolute Vina, but neither uncertainty layer resolved the contrast. Separate endpoint arms were heterogeneous: Ki was negative with a chemical-cluster interval excluding zero (-0.033, -0.069 to -0.005), whereas the much smaller Kd arm reversed sign (+0.038, -0.008 to 0.080); IC50 and EC50 were unresolved. These post hoc arms expose endpoint dependence and are not four confirmatory tests.

The old Docking-44-ChEMBL analysis is retained only as a sparse support sensitivity. It evaluated 137 ligands, 38 targets and 2,522 non-tied pairs; its residual-minus-absolute interval was -0.065 to 0.063. The two docking matrices share ChEMBL as an activity source, so they are alternative docking supports for the same observed-pair estimand rather than independent experimental replications.

Two denser full-Standard-InChIKey-matched checks narrowed chemical-sampling uncertainty but not target-composition uncertainty. DAVIS contained 56 evaluated ligands and 6,065 orderable comparisons across 21 targets; PKIS2 contained 154 ligands and 16,417 comparisons across 21 targets. Residual-minus-absolute differences were -0.009 (chemical-cluster 95% interval -0.030 to 0.010) and -0.003 (-0.018 to 0.012), while their target-jackknife intervals were -0.081 to 0.061 and -0.060 to 0.053. Post hoc margin curves are reported in Supplementary Information. Two outcome-conditioned strata reversed the aggregate direction: DAVIS both-uncensored comparisons gave 0.044 (95% chemical-cluster interval 0.004 to 0.089) and PKIS2 both-active comparisons gave 0.052 (0.002 to 0.109). Because the experimental outcomes define

stratum membership and the intervals are unadjusted for multiplicity, these positive strata are hypothesis-generating and do not replace the aggregate estimands.

The stepwise decomposition identifies exactly where rankings can change. Let μ_j and s_j be a reference target mean and residual standard deviation, $\bar{X}_i$ the evaluation-ligand row mean and $\bar{X}_{\text{ref}}$ the reference grand mean. A target-centred scaled score S_{ij} and scaled two-way residual T_{ij} satisfy

$$T_{ij} - S_{ij} = -\frac{\bar{X}_i - \bar{X}_{\text{ref}}}{s_j}. \quad (9)$$

Across the evaluation matrix this difference is the outer product of a ligand-specific scalar and the fixed inverse-scale target profile, hence a rank-one correction. It is largest for targets with small residual spread and changes sign with the ligand's row mean. Thus row subtraction alone is rank invariant (Eq. 8); unequal target scaling is the operation that turns it into a different score rule. The correlated step contrasts are post hoc and multiplicity-unadjusted and are reported in Supplementary Information as a localization of the rule difference, not as separate discoveries.

Across the broad primary and two dense checks, no tested normalized Vina representation had a higher aggregate point estimate than absolute Vina. This is a result on the measured pairs and fixed target panels, not evidence of equivalence, absence of subgroup effects or failure to retrieve targets that were never assayed.

Coverage, binding-assay, same-endpoint, activity-threshold, pair-weighted and chemical- redundancy analyses are reported in Supplementary Information. Their ordering is not uniform, reinforcing the restricted interpretation: the broad result is an observed-pair stress test under a sparse, non-random measurement process, not a reverse-docking retrieval benchmark or a general statement about selectivity.

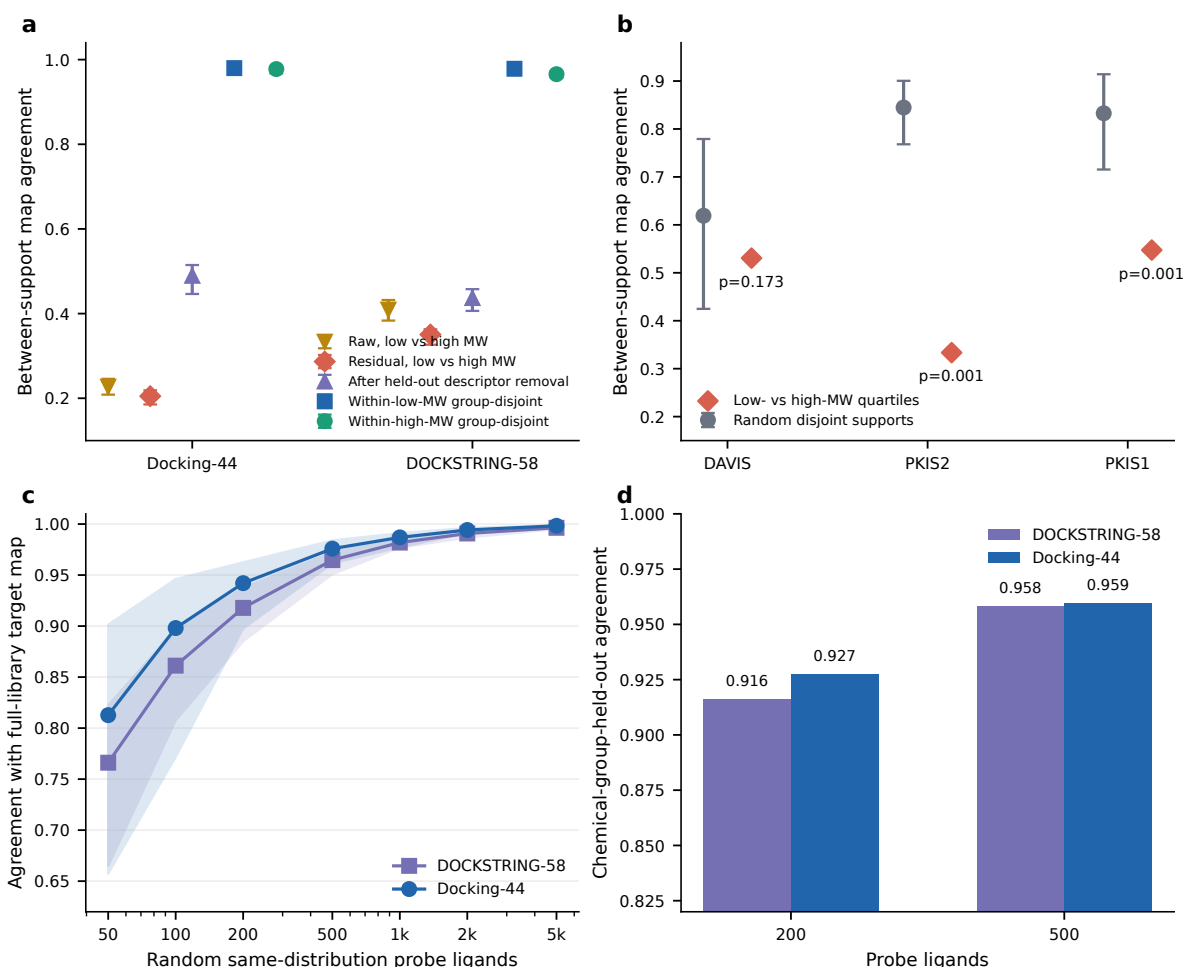


Figure 2: Exploratory molecular-size-domain shifts alter the target-pair map, whereas within-library sampling recovers it rapidly on the tested supports. **a**, Agreement of low- and high-molecular-weight docking maps on raw and row-centred surfaces and after group-held-out feature removal, compared with chemically group-disjoint splits made separately within each extreme MW band. Error bars are chemical-group bootstrap intervals or repeated-split 2.5–97.5% composition-sensitivity ranges; the latter are not population confidence intervals. **b**, Agreement between low- and high-molecular-weight experimental maps compared with random disjoint supports of the same sizes; the displayed p -values are row-randomization probabilities conditional on each observed target panel, while chemical dependence is evaluated separately through cluster resampling. **c**, Agreement of maps estimated from 50–5,000 same-distribution ligands with the full observed source-library map; shading is the 2.5–97.5% range over 100 samples. **d**, Mean agreement when the chemical groups used for estimation are excluded from the comparison map. The MW split and inspected descriptor family are post hoc; recovery curves are descriptive and support-specific, not universal sample-size thresholds.

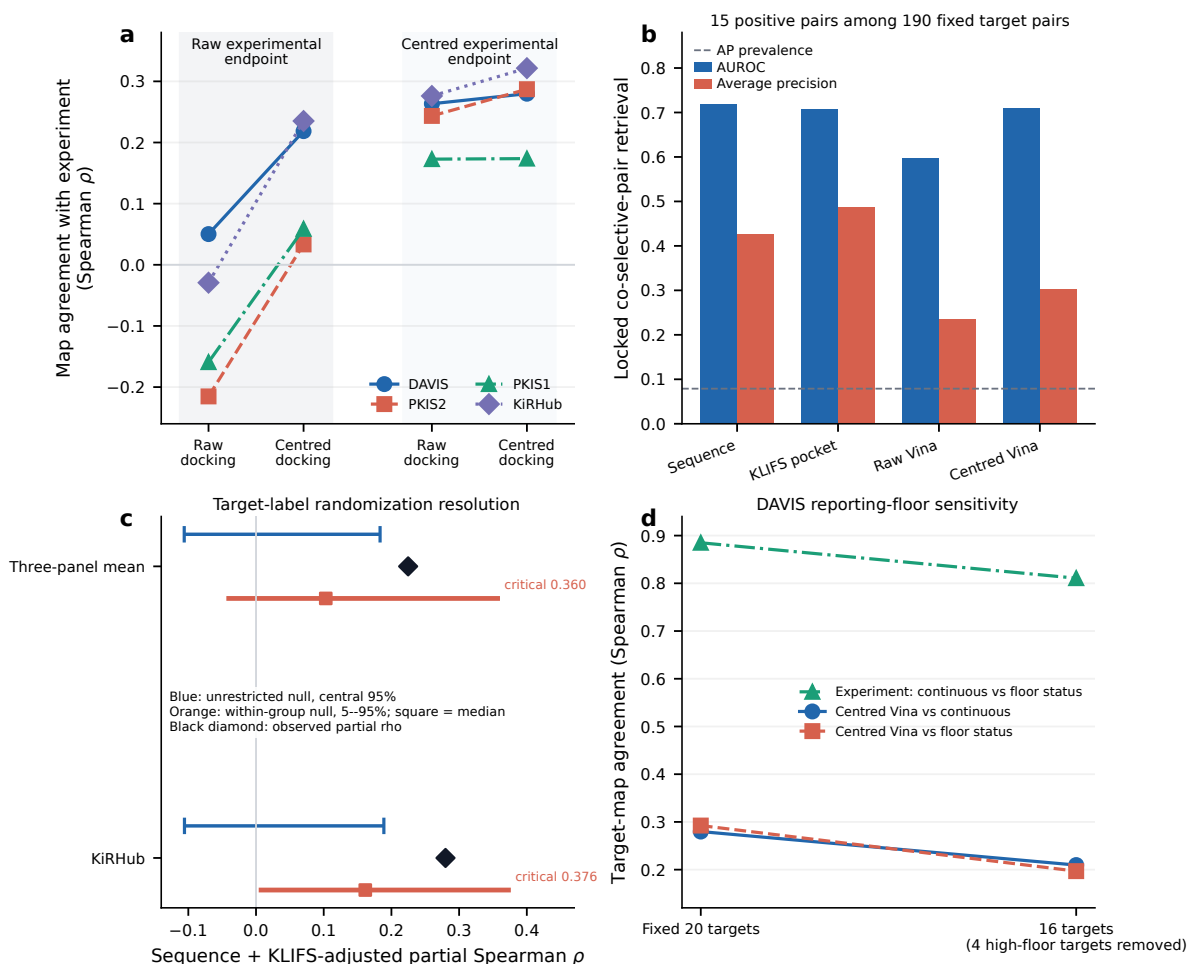


Figure 3: Experimental agreement is bounded by endpoint transformation, protein structure and measurement resolution. **a**, The complete docking-transform by experimental-endpoint factorial. Each line changes only the docking transformation within a fixed endpoint; the much larger increments occur against the raw endpoint. **b**, Retrieval of the locked 15-of-190 co-selective-pair endpoint by sequence identity, KLIFS pocket identity, raw Vina and centred Vina; the dashed line is AP prevalence. **c**, Observed sequence- and pocket-adjusted partial correlations beside unrestricted and within- KLIFS-group randomization ranges. Restricted medians and one-sided 5% critical values describe fixed-matrix resolution, not power. **d**, DAVIS comparison before and after removing four targets with more than 90% of measurements at the reporting floor.

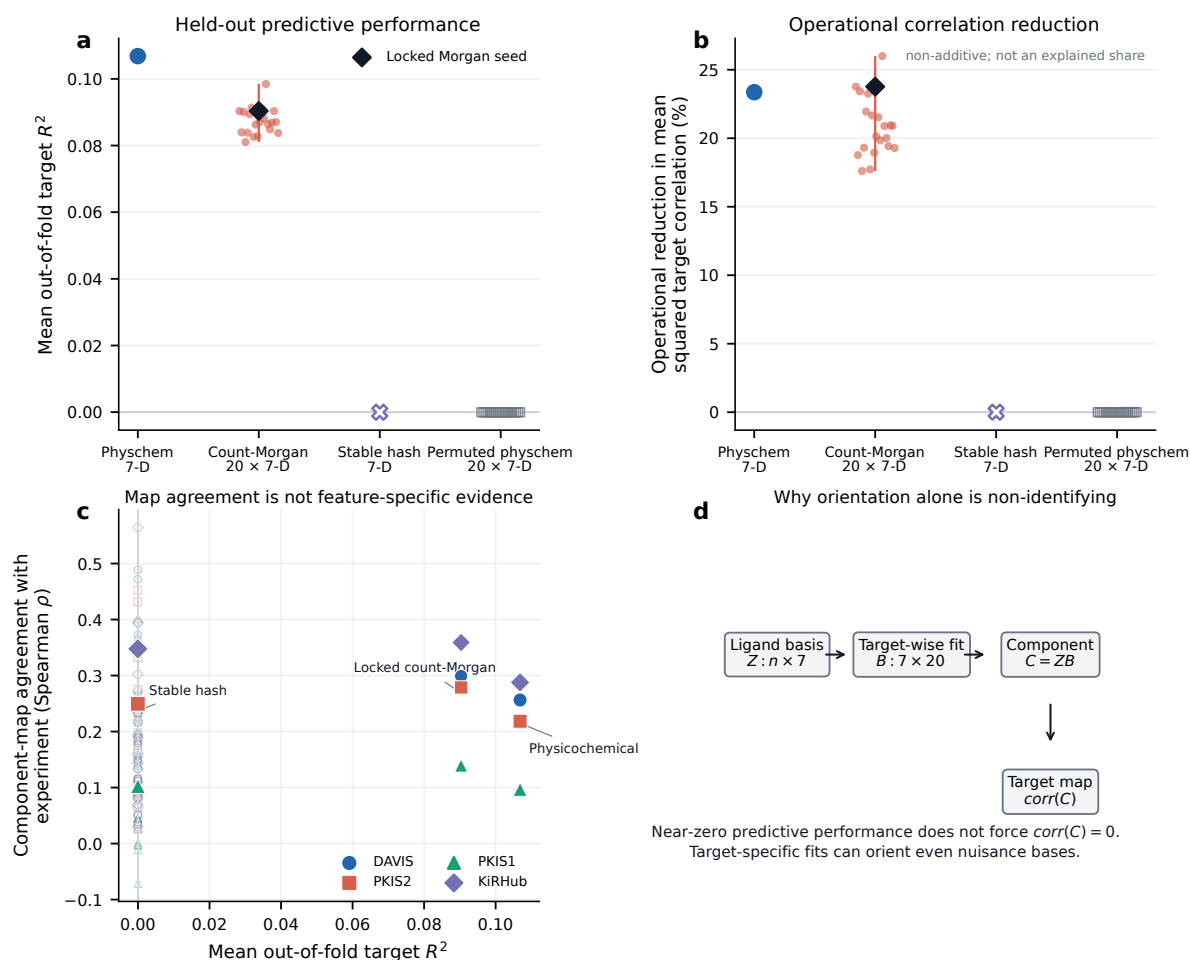


Figure 4: Held-out predictive performance and fitted-component map orientation are different estimands. **a**, Mean out-of-fold target R^2 for seven physicochemical features, 20 seven-dimensional count-Morgan projections, a stable-hash basis and 20 row- permuted physicochemical bases. The diamond marks the locked Morgan seed. **b**, Operational reduction in mean squared target correlation after subtracting each out-of-fold surface; this non-additive ratio is not an explained share. **c**, Agreement of each component map with four experimental maps versus held-out predictive performance. Nuisance bases near zero on the horizontal axis can have substantial map agreement. **d**, Target-wise coefficients orient any shared ligand basis; subsequent column correlation removes the predicted component's scale. Seed clouds are algorithmic sensitivities, not confidence intervals.

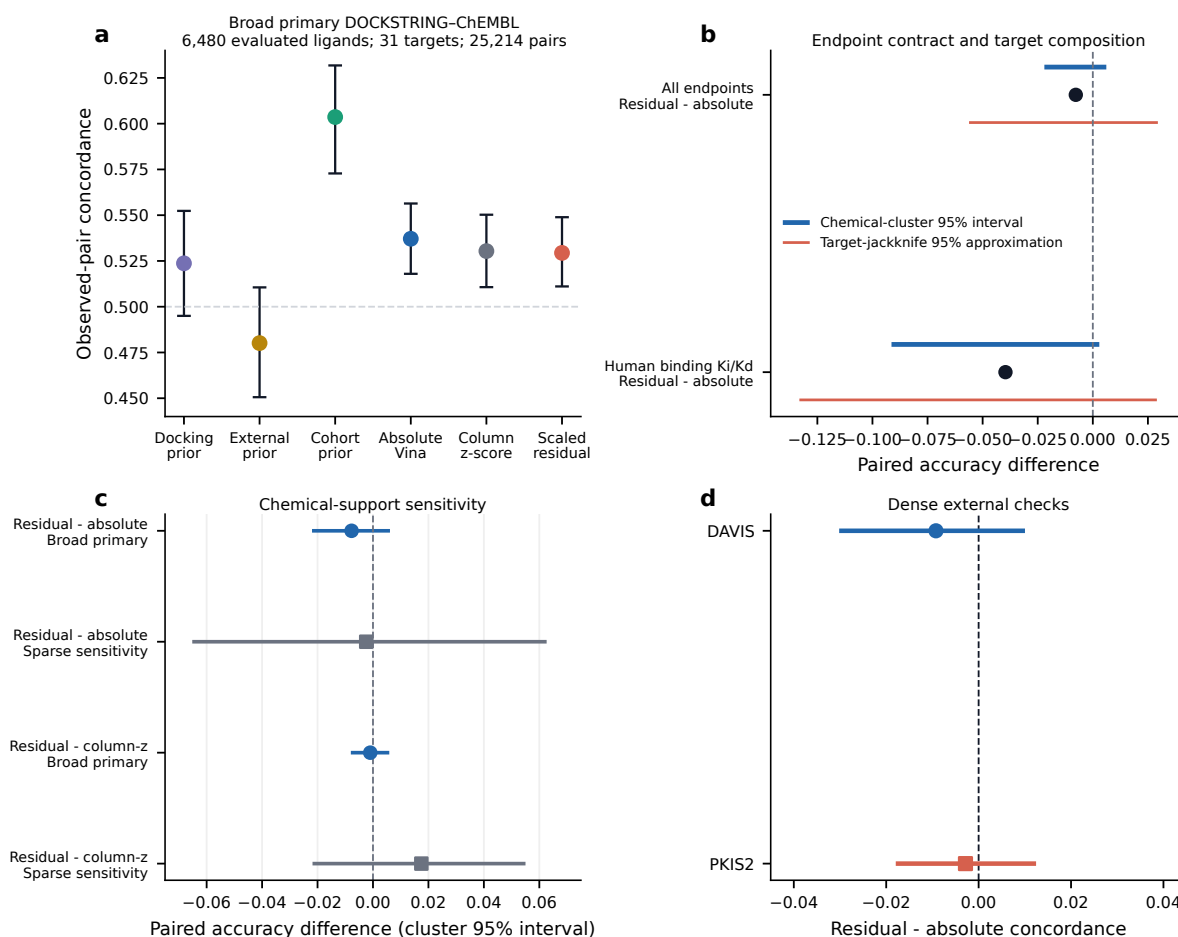


Figure 5: Within-ligand centring does not imply a target-ranking gain. **a**, Broad primary DOCKSTRING-ChEMBL observed-pair concordance. Error bars are conservative unions of Murcko- and Butina-cluster 95% intervals; the dashed line marks chance. The cohort prior is scaffold-held-out but benchmark-fitted and is not a deployable ligand-specific predictor. **b**, Residual-minus-absolute contrasts for the broad all-endpoint primary and the endpoint-restricted human binding Ki/Kd arm. Blue intervals resample chemical clusters conditional on fixed target labels; orange intervals are delete-one-target jackknife approximations. **c**, Broad primary contrasts beside the original sparse Docking-44 sensitivity. **d**, Dense full-Standard-InChIKey-matched DAVIS and PKIS2 checks; blue and orange distinguish the two datasets and both show chemical-cluster 95% intervals.

4 Discussion

The main practical result is that a docking target-correlation map is conditional on the observed source library used to estimate it, but inexpensive to remeasure within that library. In exploratory post hoc analyses, maps from the lowest and highest molecular-weight domains agreed at only 0.205 and 0.351 in the two Vina panels, whereas chemically disjoint controls with matched molecular-weight distributions agreed at 0.991 and 0.994. This gap is too large to attribute to finite sample size or chemical disjointness alone: the shift was also present on the raw surfaces, whereas MW-stratified chemical-group-disjoint halves made separately within each extreme band remained highly concordant. The largest shifts across the seven inspected descriptor splits formed a correlated molecular-size family, while other descriptor splits were weaker; this does not identify a unique causal descriptor. Conversely, 200 same-distribution ligands recovered the full-library maps at 0.942 and 0.918, with nearly unchanged agreement when the sampled chemical groups were excluded from the reference. A practitioner therefore need not dock an entire screening collection to diagnose panel redundancy in these two data sets: a representative probe set of a few hundred compounds gave high descriptive agreement on the observed supports. This does not establish 200 ligands as a universal accuracy threshold. The estimate should be repeated after a material chemical-domain shift and reported with the scoring pipeline, receptor panel and ligand support; it should not be exported as an intrinsic network of the targets.

The spectral analysis explains why this distinction matters. Both matrices contain a nearly uniform ligand-wide score axis. Within-ligand centring removes that axis and exposes a higher-dimensional, still correlated residual map; PR merely places this dependence on an effective-component scale equivalent to mean squared target correlation. Physicochemical and count-Morgan bases both predicted part of the score surface out of chemical group, so the low-dimensional response was not specific to seven named descriptors. More strikingly, a stable-hash nuisance basis had essentially zero held-out predictive performance while its fitted-component correlation map still resembled the experimental maps. Separate target coefficients inherit a random sketch of target covariance, and correlation removes the predicted columns' scale. Component-map concordance must therefore be paired with out-of-fold predictive-performance and nuisance controls. Descriptor removal also restored only part of low/high- domain agreement: molecular size is a domain marker, not an identified sole cause, and neither residual PR nor a fitted-component map is a docking-quality score.

External agreement supplies a further boundary rather than a biological certificate. A matched raw-to-centred comparison changes predictor and endpoint together. Holding the experimental endpoint fixed, the docking-side increments were small and unresolved, whereas the same docking transformation produced much larger increments against the uncentred endpoint. Sequence and KLIFS pocket similarity had comparable or higher retrieval point estimates for the fixed co-selective pairs than centred Vina, and strict within-group permutations did not resolve an additional within-family component. Because even a nonpredictive nuisance component could inherit this external geometry, decomposing the surface into a fitted component and its correlated remainder cannot localize the agreement. A docking map's concordance with an assay panel therefore describes co-response on the tested support; by itself it does not establish target-resolved recognition.

For ranking, the boundary is quantitative rather than merely cautionary. Subtracting a ligand row mean cannot reorder that ligand's targets, so no diagnostic built on the residual map is itself a ranking result. Unequal target scaling turns that otherwise invariant row term into an exact rank-one correction and therefore defines a new scoring rule. In the broad primary benchmark absolute Vina had the highest aggregate point estimate among the tested docking-derived representations, but a scaffold- held-out target-prevalence prior was higher than every ligand-specific rule. Chemical- cluster inference also did not resolve absolute Vina's small advantage over the external- docking target-only prior. Thus this sparse observed-pair benchmark did not establish that the ligand-specific Vina scores added ordering information beyond target prevalence. Chemical- sampling intervals for residual-minus-absolute scores were narrow, whereas target-jackknife sensitivities were much wider. The less heterogeneous, endpoint-restricted human binding K_i/K_d arm also favoured absolute Vina in point estimate, but both uncertainty layers included zero. That arm can still mix K_i - and K_d -derived cells within a pair, and the separate K_i and K_d arms differed in sign. This endpoint dependence precludes a universal winner. Normalization-path contrasts and the margin curves retained in Supplementary Information are post hoc localizations, not confirmatory evidence.

This task dependence is consistent with earlier benchmarking. MASC did not generally improve enrichment across eight targets and ten scoring functions; multiple-target screening showed that MASC-like decisions depend on the reference receptor set; and heavy-atom normalization improved most tested GOLD panels while impairing most Glide panels [7, 9, 23]. Normalization is therefore a task-specific scoring rule, not a universally beneficial correction. The separation

the audit enforces—between a co-response geometry over targets and a decision about one compound—is the same separation that CASF draws between the scoring, ranking and screening powers of a function [14], and the target-pair estimand is the descriptive counterpart of the ligand–target interaction terms that proteochemometric models fit explicitly [26,27].

Both large matrices use Vina-family scores. In a secondary fixed-pose sensitivity, five of six non-Vina functions met weak shared-axis criteria on Vina-selected poses, while PLECScore falsified a universal-threshold claim. Pose selection remained Vina dependent and several functions share training data. Moreover, a broadly positive score matrix can yield a nearly uniform leading vector under these weak criteria. This is neither an independent replication of a complete docking pipeline nor evidence of scorer-generality.

The remaining limits are as follows. The experimental comparison is restricted to 20 kinases and observed target pairs; the target panels are not random samples from protein space. DAVIS, PKIS2, PKIS1 and KiRHub differ in assay technology, compound libraries and measurement scale. DAVIS has 67.0% of fixed-panel values at its pKd floor, and its continuous target map largely tracks binary above-floor status ($\rho = 0.885$ on the fixed 20-target panel). Its sensitivity therefore supports coarse co-response or measurement-limit status more strongly than fine quantitative affinity geometry; KiRHub has 20.2% exact residual-activity values at 100 and lacks structures needed for a complete chemical-overlap audit. The structural controls are post hoc, and exchangeability within KLIFS groups is itself an approximation. The small-panel recovery experiment uses same-distribution samples and its high agreement is descriptive for the two observed source libraries, not a universal 200-ligand accuracy threshold. The broad ligand-wise ChEMBL benchmark is non-randomly observed and assay heterogeneous; its smaller human-binding Ki/Kd arm is less heterogeneous but can still compare Ki- and Kd-derived cells. The dense DAVIS block is strongly floor-censored, PKIS2 is a single-dose inhibition panel, and exact matching to DOCKSTRING selects restricted chemical subsets. Chemical intervals are conditional on 21 fixed targets in the dense checks and 31 fixed targets in the broad ChEMBL analysis; the original sparse Docking-44 sensitivity uses 38 targets. Target jackknives are panel-composition sensitivities rather than population intervals. Finally, neither PR nor external target-pair concordance assesses docking poses or absolute affinity accuracy.

The reusable audit follows directly from these boundaries: state the score and endpoint estimands; report both uncentred and within-ligand-centred correlation geometry; express PR beside mean squared correlation and the full spectrum; use transformation-matched nulls; quantify

Table 3: Decision-oriented audit for a ligand-by-target score matrix. The steps separate a descriptive matrix diagnosis from claims about biological geometry or compound- level target ranking.

Stage	Required comparison	Decision boundary
Define the estimand	Name the score transformation, experimental endpoint, ligand support and target panel separately.	Do not attribute a joint change to the predictor when predictor and endpoint both change.
Diagnose dependence	Show raw and residual correlation spectra, covariance sensitivity, PR beside mean r^2 , the full scree and transformation-matched nulls.	Use the result to detect a shared ligand axis or residual redundancy, never as a stand-alone docking-quality score.
Stress transport	Vary missing-data handling, target composition and chemical support; compare descriptive same-distribution recovery with exploratory molecular-size-domain shifts and matched controls.	If domain drift exceeds resampling error, report and re-estimate a domain-conditional map rather than exporting one universal network; do not treat an observed recovery sample size as a universal threshold.
Test target-pair fidelity	Evaluate the full docking $\times$ experiment transformation factorial against a fixed endpoint; include sequence, pocket and fitted-component nuisance baselines.	For any fitted component, report held-out predictive performance beside its correlation map; claim target-pair information only beyond the stated controls.
Test the score rule	Establish rank invariances, compare ligand-invariant target priors and evaluate every offset, scale and ordering step on dense observed target profiles.	Deploy a normalization for ranking only after a paired task-level gain with chemical- and target-aware uncertainty.

chemical and target-panel sensitivity; validate target-pair geometry and ligand-wise ranking separately; pair any fitted-component map with held-out predictive performance and a nuisance basis; and report the full predictor-by-endpoint table whenever both sides are normalized. The released generic command-line tool implements the spectral portion; dataset-specific scripts implement the external-geometry and ranking steps. Table 3 makes the associated decisions explicit.

5 Conclusions

A Vina target-correlation map is a property of a scoring pipeline, receptor panel and ligand library together. In exploratory post hoc analyses within the observed source libraries, it changed sharply between molecular-weight-defined domains, with the strongest inspected shifts belonging to a correlated molecular-size descriptor family. Descriptive sampling experiments gave high same-domain recovery from 200–500 representative ligands in these two data sets; that range is not a universal accuracy threshold. The actionable protocol is therefore to estimate the map on a small probe set drawn from the intended library, use it to diagnose redundant target responses,

and re-estimate it after domain shift. Within-ligand centring exposes the residual map but does not validate it: a low-dimensional ligand-feature basis predicts part of its score structure, yet even a nonpredictive nuisance basis can show nonzero component-map agreement after correlation removes the predicted columns' scale. Component-map agreement alone is therefore non-identifying; assay transformation and protein-family structure impose additional interpretive limits.

The map is also not a compound-level target-ranking rule. Subtracting a ligand's row mean cannot change its target order; unequal target scales turn that invariant subtraction into an exact rank-one correction that must be assessed as a new score rule. On the broad observed-pair benchmark, absolute Vina had the highest aggregate point estimate among the tested docking-derived representations, while chemical-sampling and target-composition analyses did not resolve a gain for scaled residual scores; nor did the paired chemical-cluster interval resolve its small advantage over a target-only prior fitted on the external DOCKSTRING reference. Spectral diagnosis, biological validation and ligand-wise ranking are thus three separate tasks: the first can be performed cheaply, but neither of the latter two follows from an increase in residual effective dimension.

List of abbreviations

AP, average precision; AUROC, area under the receiver-operating-characteristic curve; KLIFS, Kinase–Ligand Interaction Fingerprints and Structures; OAS, Oracle Approximating Shrinkage; PC, principal component; PR, participation ratio; QAP, quadratic assignment procedure; SPD, Secondary Pharmacology Database.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The source Docking-44 dataset is available with Nikitin et al.; DOCKSTRING scores, poses, receptors and configurations are available from the DOCKSTRING release [3, 4]. The CC0 DAVIS-Complete V3 table is available from Harvard Dataverse; PKIS1, PKIS2, KiRHub, SPD, KLIFS and ChEMBL are available from their cited source repositories subject to their stated licenses [32]. The analysis code, frozen redistributable inputs, checksum manifests and derived results are available in the public GitHub repository and its Zenodo archive [52, 53]. The release corresponding to this article is identified by version and date in the repository's CITATION.cff; readers should cite that version-specific archive rather than the concept record, because earlier versions correspond to earlier analyses. Source files that cannot be redistributed are obtained from their cited repositories under checksum-verified retrieval contracts documented in the package; they are not downloaded by the default report-layer build. KiRHub source tables are not redistributed. The associated article is CC BY-NC-ND 4.0, while the official KiRHub portal separately identifies the underlying kinase-inhibition data as the property of Reaction Biology Corporation and requests proper acknowledgement. We do not assert a separate license for those data; the package contains aggregate analytical summaries, not compound-by-target activities or per-compound profiles.

Every quantity this article reports for the fixed 20-kinase panel is released. The complete 190-edge target-pair ledger—raw and within-ligand-centred geometry for all four experimental panels and for the docking reference, together with the locked 15-pair endpoint—is distributed with the code, and it regenerates the full predictor-by-endpoint factorial, the cross-panel agreement summary and every retrieval metric reported here. That ledger is an aggregate statistic rather than a redistribution: each entry is one correlation coefficient computed across compounds, not a compound-level value or profile. No conclusion in this article rests on data that a reader cannot obtain: the license-restricted KiRHub and PKIS1 sources are retrieved by checksum-verified scripts from their publishers, and every other input is redistributed with the package.

Competing interests

The authors declare no financial or commercial competing interests. Two authors contributed to the source study of the Docking-44 matrix [4]; this relationship is disclosed because that dataset is reanalysed here, and the chemical and receptor overlap between the two score matrices is

1097 quantified in the Results.

1098 **Funding**

1099 This research received no specific grant from any funding agency in the public, commercial, or
1100 not-for-profit sectors.

1101 **Authors' contributions**

1102 A.L. conceived the study, performed the analyses and drafted the manuscript. V.S. and M.F.
1103 contributed to study design and interpretation. All authors reviewed the manuscript and are
1104 responsible for the final submitted version.

1105 **Acknowledgements**

1106 The authors thank the developers and maintainers of the open datasets and software used in this
1107 study. The authors acknowledge Reaction Biology Corporation as the generator and identified
1108 owner of the kinase-inhibition data accessed through the KIRHub publication and portal.

1109 **Authors' information**

1110 ORCID: Adeliya Leleytner, 0009-0001-8214-2496; Victor Safronov, 0009-0003-5668-0201; Maxim
1111 Fedorov, 0000-0003-3901-3565.

1112 **References**

- 1113 [1] Trott O, Olson AJ (2010) AutoDock Vina: improving the speed and accuracy of docking with a
1114 new scoring function, efficient optimization, and multithreading. *J Comput Chem* 31:455–461.
1115 doi:10.1002/jcc.21334.
- 1116 [2] Ding J, Tang S, Mei Z et al (2023) Vina-GPU 2.0: further accelerating AutoDock Vina
1117 and its derivatives with graphics processing units. *J Chem Inf Model* 63:1982–1998.
1118 doi:10.1021/acs.jcim.2c01504.
- 1119 [3] García-Ortegón M, Simm GNC, Tripp AJ, Hernández-Lobato JM, Bender A, Bacallado S (2022)
1120 DOCKSTRING: easy molecular docking yields better benchmarks for ligand design. *J Chem Inf*
1121 *Model* 62:3486–3502. doi:10.1021/acs.jcim.1c01334.

- [4] Nikitin I, Morgunov I, Safronov V, Kalyuzhnaya A, Fedorov M (2025) Towards explainable computational toxicology: linking antitargets to rodent acute toxicity. *Pharmaceutics* 17:1573. doi:10.3390/pharmaceutics17121573.
- [5] Pan Y, Huang N, Cho S, MacKerell AD Jr (2003) Consideration of molecular weight during compound selection in virtual target-based database screening. *J Chem Inf Comput Sci* 43:267–272. doi:10.1021/ci020055f.
- [6] Vigers GPA, Rizzi JP (2004) Multiple active site corrections for docking and virtual screening. *J Med Chem* 47:80–89. doi:10.1021/jm030161o.
- [7] Jacobsson M, Karlén A (2006) Ligand bias of scoring functions in structure-based virtual screening. *J Chem Inf Model* 46:1334–1343. doi:10.1021/ci050407t.
- [8] Carta G, Knox AJS, Lloyd DG (2007) Unbiasing scoring functions: a new normalization and rescoring strategy. *J Chem Inf Model* 47:1564–1571. doi:10.1021/ci600471m.
- [9] Ibrahim TM, Bauer MR, Boeckler FM (2015) Applying DEKOIS 2.0 in structure-based virtual screening to probe the impact of preparation procedures and score normalization. *J Cheminform* 7:21. doi:10.1186/s13321-015-0074-6.
- [10] Sieg J, Flachsenberg F, Rarey M (2019) In need of bias control: evaluating chemical data for machine learning in structure-based virtual screening. *J Chem Inf Model* 59:947–961. doi:10.1021/acs.jcim.8b00712.
- [11] Wallach I, Heifets A (2018) Most ligand-based classification benchmarks reward memorization rather than generalization. *J Chem Inf Model* 58:916–932. doi:10.1021/acs.jcim.7b00403.
- [12] Chen L, Cruz A, Ramsey S, Dickson CJ, Duca JS, Hornak V, Koes DR, Kurtzman T (2019) Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PLoS One* 14:e0220113. doi:10.1371/journal.pone.0220113.
- [13] Gabel J, Desaphy J, Rognan D (2014) Beware of machine learning-based scoring functions—on the danger of developing black boxes. *J Chem Inf Model* 54:2807–2815. doi:10.1021/ci500406k.
- [14] Su M, Yang Q, Du Y, Feng G, Liu Z, Li Y, Wang R (2019) Comparative assessment of scoring functions: the CASF-2016 update. *J Chem Inf Model* 59:895–913. doi:10.1021/acs.jcim.8b00545.
- [15] Luo Q, Zhao L, Hu J, Jin H, Liu Z, Zhang L (2017) The scoring bias in reverse docking and the score normalization strategy to improve success rate of target fishing. *PLoS One* 12:e0171433. doi:10.1371/journal.pone.0171433.

- 1152 [16] Lapillo M, Tuccinardi T, Martinelli A, Macchia M, Giordano A, Poli G (2019) Extensive reliability eval-
1153 uation of docking-based target-fishing strategies. *Int J Mol Sci* 20:1023. doi:10.3390/ijms20051023.
- 1154 [17] Lee A, Kim D (2020) CRDS: consensus reverse docking system for target fishing. *Bioinformatics*
1155 36:959–960. doi:10.1093/bioinformatics/btz656.
- 1156 [18] Yang L, Luo H, Chen J, Xing Q, He L (2009) SePreSA: a server for the prediction of populations
1157 susceptible to serious adverse drug reactions implementing the methodology of a chemical–protein
1158 interactome. *Nucleic Acids Res* 37:W406–W412. doi:10.1093/nar/gkp312.
- 1159 [19] Casey FP, Pihan E, Shields DC (2009) Discovery of small molecule inhibitors of protein–protein
1160 interactions using combined ligand and target score normalization. *J Chem Inf Model* 49:2708–2717.
1161 doi:10.1021/ci900294x.
- 1162 [20] Yang L, Chen J, Shi L, Hudock MP, Wang K, He L (2010) Identifying unexpected therapeutic targets
1163 via chemical–protein interactome. *PLoS One* 5:e9568. doi:10.1371/journal.pone.0009568.
- 1164 [21] Luo H, Chen J, Shi L, Mikailov M, Zhu H, Wang K, He L, Yang L (2011) DRAR-CPI: a server
1165 for identifying drug repositioning potential and adverse drug reactions via the chemical–protein
1166 interactome. *Nucleic Acids Res* 39:W492–W498. doi:10.1093/nar/gkr299.
- 1167 [22] Fukunishi Y, Mikami Y, Nakamura H (2005) Similarities among receptor pockets and among
1168 compounds: analysis and application to in silico ligand screening. *J Mol Graph Model* 24:34–45.
1169 doi:10.1016/j.jmgm.2005.04.004.
- 1170 [23] Fukunishi Y, Mikami Y, Kubota S, Nakamura H (2006) Multiple target screening method
1171 for robust and accurate in silico ligand screening. *J Mol Graph Model* 25:61–70.
1172 doi:10.1016/j.jmgm.2005.11.006.
- 1173 [24] Fukunishi Y, Mikami Y, Takedomi K, Yamanouchi M, Shima H, Nakamura H (2006) Classification of
1174 chemical compounds by protein–compound docking for use in designing a focused library. *J Med*
1175 *Chem* 49:523–533. doi:10.1021/jm050480a.
- 1176 [25] Fukunishi Y, Yamashita Y, Mashimo T, Nakamura H (2018) Prediction of protein–compound binding
1177 energies from known activity data: docking-score- based method and its applications. *Mol Inform*
1178 37:e1700120. doi:10.1002/minf.201700120.
- 1179 [26] Cortés-Ciriano I, Ain QU, Subramanian V et al (2015) Polypharmacology modelling using pro-
1180 teochemometrics (PCM): recent methodological developments, applications to target families, and
1181 future prospects. *MedChemComm* 6:24–50. doi:10.1039/C4MD00216D.

- 1182 [27] van Westen GJP, Wegner JK, IJzerman AP, van Vlijmen HWT, Bender A (2011) Proteochemo-
1183 metric modeling as a tool to design selective compounds and for extrapolating to novel targets.
1184 *MedChemComm* 2:16–30. doi:10.1039/C0MD00165A.
- 1185 [28] Lee AA, Brenner MP, Colwell LJ (2016) Predicting protein–ligand affinity with a random matrix
1186 framework. *Proc Natl Acad Sci USA* 113:13564–13569. doi:10.1073/pnas.1611138113.
- 1187 [29] Graczyk PP (2007) Gini coefficient: a new way to express selectivity of kinase inhibitors against a
1188 family of kinases. *J Med Chem* 50:5773–5779. doi:10.1021/jm070562u.
- 1189 [30] Bosc N, Wroblowski B, Meyer C, Bonnet P (2017) The use of novel selectivity metrics in kinase
1190 research. *BMC Bioinformatics* 18:17. doi:10.1186/s12859-016-1413-y.
- 1191 [31] Davis MI, Hunt JP, Herrgard S et al (2011) Comprehensive analysis of kinase inhibitor selectivity.
1192 *Nat Biotechnol* 29:1046–1051. doi:10.1038/nbt.1990.
- 1193 [32] Wu MH (2025) A Complete and Modification-Aware DAVIS Dataset. Harvard Dataverse, V3.
1194 doi:10.7910/DVN/RTQGP1.
- 1195 [33] Elkins JM, Fedele V, Szklarz M et al (2016) Comprehensive characterization of the Published Kinase
1196 Inhibitor Set. *Nat Biotechnol* 34:95–103. doi:10.1038/nbt.3374.
- 1197 [34] Drewry DH, Wells CI, Andrews DM et al (2017) Progress towards a public chemoge-
1198 nomic set for protein kinases and a call for contributions. *PLoS One* 12:e0181585.
1199 doi:10.1371/journal.pone.0181585.
- 1200 [35] Saifudeen M, Zhu S, Liang S et al (2026) Comprehensive profiling of clinically approved kinase
1201 inhibitors reveals mutation-specific inhibitors and opportunities for drug repurposing. *Nat Biotechnol*.
1202 doi:10.1038/s41587-026-03090-8.
- 1203 [36] Sutherland JJ, Yonchev D, Fekete A, Urban L (2023) A preclinical secondary pharmacology resource
1204 illuminates target–adverse drug reaction associations of marketed drugs. *Nat Commun* 14:4323.
1205 doi:10.1038/s41467-023-40064-9.
- 1206 [37] van Linden OPJ, Kooistra AJ, Leurs R, de Esch IJP, de Graaf C (2014) KLIFS: a knowledge-
1207 based structural database to navigate kinase–ligand interaction space. *J Med Chem* 57:249–277.
1208 doi:10.1021/jm400378w.
- 1209 [38] Kooistra AJ, Kanev GK, van Linden OPJ, Leurs R, de Esch IJP, de Graaf C (2016) KLIFS: a structural
1210 kinase–ligand interaction database. *Nucleic Acids Res* 44:D365–D371. doi:10.1093/nar/gkv1082.
- 1211 [39] Zdrazil B, Felix E, Hunter F et al (2024) The ChEMBL Database in 2023: a drug discovery platform
1212 spanning multiple bioactivity data types and time periods. *Nucleic Acids Res* 52:D1180–D1192.
1213 doi:10.1093/nar/gkad1004.

- 1214 [40] ChEMBL Team (2024) ChEMBL database release 34. doi:10.6019/CHEMBL.database.34.
- 1215 [41] Mazzucato L, Fontanini A, La Camera G (2016) Stimuli reduce the dimensionality of cortical activity.
1216 Front Syst Neurosci 10:11. doi:10.3389/fnsys.2016.00011.
- 1217 [42] Roy O, Vetterli M (2007) The effective rank: a measure of effective dimensionality. In: 15th European
1218 Signal Processing Conference, Poznań, pp 606–610. doi:10.5281/zenodo.40328.
- 1219 [43] International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human
1220 Use (2019) ICH E9(R1) addendum on estimands and sensitivity analysis in clinical trials to the
1221 guideline on statistical principles for clinical trials. [https://database.ich.org/sites/default/](https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf)
1222 [files/E9-R1_Step4_Guideline_2019_1203.pdf](https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf).
- 1223 [44] Hubert L, Schultz J (1976) Quadratic assignment as a general data analysis strategy. Br J Math Stat
1224 Psychol 29:190–241. doi:10.1111/j.2044-8317.1976.tb00714.x.
- 1225 [45] Dekker D, Krackhardt D, Snijders TAB (2007) Sensitivity of MRQAP tests to collinearity and autocor-
1226 relation conditions. Psychometrika 72:563–581. doi:10.1007/s11336-007-9016-1.
- 1227 [46] Bemis GW, Murcko MA (1996) The properties of known drugs. 1. Molecular frameworks. J Med
1228 Chem 39:2887–2893. doi:10.1021/jm9602928.
- 1229 [47] Morgan HL (1965) The generation of a unique machine description for chemical structures—a tech-
1230 nique developed at Chemical Abstracts Service. J Chem Doc 5:107–113. doi:10.1021/c160017a018.
- 1231 [48] Rogers D, Hahn M (2010) Extended-connectivity fingerprints. J Chem Inf Model 50:742–754.
1232 doi:10.1021/ci100050t.
- 1233 [49] Butina D (1999) Unsupervised data base clustering based on Daylight’s fingerprint and Tanimoto
1234 similarity: a fast and automated way to cluster small and large data sets. J Chem Inf Comput Sci
1235 39:747–750. doi:10.1021/ci9803381.
- 1236 [50] Landrum G et al (2025) RDKit: open-source cheminformatics software, release 2025.03.6. Zenodo.
1237 doi:10.5281/zenodo.16996017.
- 1238 [51] Chen Y, Wiesel A, Eldar YC, Hero AO (2010) Shrinkage algorithms for MMSE covariance estimation.
1239 IEEE Trans Signal Process 58:5016–5029. doi:10.1109/TSP.2010.2053029.
- 1240 [52] Leleytner A, Safronov V, Fedorov M (2026) Target-correlation maps are library-conditional but recov-
1241 erable from a few hundred ligands in two large Vina panels: code, frozen data and derived evidence.
1242 GitHub. <https://github.com/AdeliyaLeleytner/vina-score-spectral-concentration>.

1243 [53] Leleytner A, Safronov V, Fedorov M (2026) Target-correlation maps are library-conditional but
1244 recoverable from a few hundred ligands in two large Vina panels: reproducibility archive. Zenodo,
1245 version 3.0.0. doi:10.5281/zenodo.21865607.